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Advances and challenges in cryogenic liquefied hydrogen (LH2) maritime transportation: Thermal performance, tank design, boil-off gas management, and techno-economic considerations

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ABSTRACT

The importance of maritime liquid hydrogen (LH2) transportation is increasing rapidly to achieve a future global carbon-neutral energy trade. But, the extreme thermo-physical properties of LH2 introduce major engineering constraints for design and operation. Key challenges, such as boil-off gas (BOG), self-pressurization, cooling demand, and high-cost due to the need of complex and optimized cryogenic tank insulation, significantly impacts LH2 transportation efficiency. This review systematically synthesizes the current state of LH2 carrier technologies. It covers liquid energy career benchmarking, tank architecture, cool down strategies of tanks, thermodynamic modeling, BOG management, and techno-economic feasibility, which have not been accumulated in any of the recent reviews. Comparative analysis confirms that LH2 exhibits the highest BOG losses (~3.44% per day) and transportation cost (~3.74 \$/GJ) among major hydrogen carriers. These happen due to the low volumetric density of hydrogen and severe heat ingress into the tanks. Single-node equilibrium assumptions can underpredict internal wall energy by up to 60% at low fill levels (≤5%). Conversely, validated multi-zone non-equilibrium models show that increased tank diameter can reduce relative BOG losses from ~1.8% to ~0.2% per day. Thermal management studies also indicate that controlled tank warming during ballast voyages can reduce heat accumulation by 41.6-54.3%. Several significant research gaps such as sloshing-coupled thermodynamics, full-scale experimental validation, insulation-material optimization, heel-level tuning, and integrated techno-economic assessment under carbon pricing scenarios are identified in this review. Co-optimization of thermodynamics, operation, naval architecture, and onboard propulsion is the core finding from the key insights of this review, which is necessary to achieve energy-efficient, safe, and cost-effective LH2 shipping. To establish LH2 maritime transport as a viable media of future global green-energy logistics, these scientific and engineering challenges need to be addressed.

ABBREVIATIONS

Acronym	Description
LH2	Liquefied Hydrogen
BOG	Boil-Off Gas
LNG	Liquefied Natural Gas
LOHC	Liquid Organic Hydrogen Carriers
MLI	Multi-Layer Insulation
PUF	Polyurethane Foam
MAWP	Maximum Allowable Working Pressure
TMZM	Thermal Multi-Zone Model
MHTB	Multi-Purpose Hydrogen Test Bed
HDR	Heat Distribution Ratio
CAPEX	Capital Expenditure
NPV	Net Present Value
GH2	Gaseous Hydrogen
DME	Dimethyl Ether
COGAS	Combined Gas and Steam
MGO	Marine Gas Oil
SCC	Social Carbon Cost
VLE	Vapor–Liquid Equilibrium
CFD	Computational Fluid Dynamics
IMO	International Maritime Organization

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1. Introduction

The increasing demand for long-distance energy transport in an efficient and sustainable way is a major concern for global trade. The urgency of de-carbonization has been intensified throughout couple of decades which has intensified the focus to use hydrogen as a clean and sustainable energy carrier. Among various storage and transport forms, liquefied hydrogen (LH2) has emerged as a promising option for large-scale and long-distance maritime energy transport. These have been intensified due to its high volumetric density and zero-carbon combustion profile. The regions with limited renewable energy sources would be greatly affected by this form of energy transport as it enables bulk energy delivery globally. This LH2 transportation technology also supports the development of a global hydrogen economy.

The transportation of the energy of natural gas has been historically dominated either through pipelines in gaseous form or as liquefied natural gas (LNG) for long distance shipping. However, the use of pipelines for energy transport becomes infeasible both in economically and technically for distances beyond roughly 2000 km. On the other hand, LNG offers better feasibility for long-distance shipping due to its superior volumetric efficiency. LNG provides about a 600-fold reduction in volume compared to its gaseous form [1–3]. However, LNG transport faces several notable drawbacks. The most significant ones include its low boiling temperature (−162 °C) which contribute to boil-off gas (BOG) losses, and carbon content that creates greenhouse gas emissions [4,5]. These limitations have encouraged the researchers to explore alternative solutions which offer carbon-free carriers. They proposed other ways to transport the energy of natural gas by other liquefied natural form such as ammonia, methanol, and hydrogen. These forms of transportation can mitigate boil-off gas (BOG) formation and reduce environmental impact [6].

Based on the advantages and challenges, and decades of experience in LNG shipping, researchers have recognized LH2 transport by sea as the most promising large-scale method for intercontinental hydrogen trade. LH2 reduces its volume by nearly three orders of magnitude to 70.8 kg/m^3. This feature makes maritime export feasible where pipelines or high-pressure gaseous transport are impractical [7]. Several pilot and design efforts have been conducted in recent years. These include the HESC Suiso Frontier demonstration voyage and classification society approvals for very large LH2 carrier designs. These steps have underscored the tangible momentum toward commercial LH2 shipping [8–13].

Although LH2 transportation offers significant advantages, it introduces critical challenges which range from thermodynamic and operational constraints to economical perspectives. Low boiling temperature of LH2 (−253 °C) is the root cause of most of the challenges that demands high-performance insulation, vacuum creation, and optimized structural materials of stable mechanical properties under cryogenic conditions [8,14,15]. Although proper insulation based on recent developments is ensured, the significant low boiling temperature of LH2 causes unavoidable heat ingress into the tanks. This phenomenon produces BOG as a continuous evaporation loss which reduces delivered cargo capacity and drives tank pressurization [16,17]. Additionally, large LH2 tanks offer relatively low maximum allowable working pressure (MAWP), and further complicate BOG retention. This often forces controlled venting unless re-liquefaction or utilization options are available [18,19].

Operationally, ship-borne LH2 tanks experience repeated loading-unloading cycle. It starts with loading, then goes to laden voyage, then faces unloading, and finally ballast (return) voyage. This cycle differs significantly from typical static, ground storage of LH2. After unloading, LH2 carriers usually hold a small amount of liquid which is known as heel with an aim to maintaining the tank's low temperature. During the return or ballast voyage, the tank gradually warms up as vapor and wall temperature rise. Typical voyages and the resulting phenomena may last for several weeks. Over time, this warming can cause vapor stratification in the ullage space. It influences tank pressurization behavior and increases pre-cooling demand before the next loading operation [18–21]. For LNG carriers, pre-cool down practices with intermittent heel spraying (8-12 h chill down) are well known and influence turnaround time. But, in the case of LH2, tanks differ in insulation type, inner-wall thickness, and cryogenic temperature, thus making it difficult to transfer operational paradigms of LNG to LH2 directly [22–25].

From modeling perspective, the accurate prediction of several key performance parameters is mandatory for tank and ship design. These include self-pressurization, boil-off rates, and cool down behavior, offering practical challenges. Different models have been proposed for better thermodynamic performance of LH2 transportation. Reduced order (lumped-mass/two-node) models provide computationally efficient tools for predicting long-duration behaviors. Although, these models have been validated against experimental test beds (e.g., NASA Multi-Purpose Hydrogen Test Bed (MHTB)), they face limitations at low fill levels as vapor thermal stratification and transient heat transfer become dominant at this case [26–29]. Experimental studies and multi-node or non-equilibrium models have demonstrated that isothermal assumptions under-predict pressure rise and misinterpret interface heat transfer. Along with these, varying tuning coefficients and heat transfer correlation across studies create model uncertainty [28–31]. Additionally, several under-resolved factors including the effect of tank thermal mass, inner-wall conduction, partitioning of heat leakage between vapor and liquid exist [32,33].

Economically, LH2 carriers offer higher transport cost due to BOG losses, insulation CAPEX, and the cost of liquefaction. Studies show that this cost is comparatively higher than for alternative carriers such as ammonia, methanol, dimethyl ether (DME) etc. This is primarily due to the greater BOG losses and higher storage costs, which are sensitive to voyage distance, atmospheric temperature, liquefaction cost, and carbon pricing [34–39]. However, strategic utilization of BOG can materially affect the vessel's net present value and optimal design choices. BOG has the potential to be used as onboard fuel and to be adopted for re-liquefaction or recovery systems [40]. Again, techno-economic analyses demonstrate that co-optimizing tank maximum allowable working pressure (MAWP) and cruising speed can alter net present value (NPV) outcomes by hundreds of millions USD. This is possible under plausible fuel-price and carbon-price scenarios [41,42].

Several research gaps thus emerge at the intersection of thermodynamics, operations and economics. The first one is the modeling fidelity at low heel percentages which includes vapor stratification, wall thermal inertia, and their impact on pressurization. The most reduced-order models need consistent calibration and analytical correction factors [43–45]. Secondly, experimental data are very limited, and the only operational sea trial (Suiso Frontier) used a small Type-C tank. But it may not represent future large carriers. So, scalability and upscaling evidence for thousands-to hundreds-of-thousands-m^3 LH2 tanks are needed. Integrated ship system studies and practical methods for BOG management are other areas of research gaps. But very few studies have worked with accumulating all of the challenges and research gaps of hydrogen transportation. Most of the studies of energy transportation are based on LNG carriers. Either they have worked with a specific technical area of LNG storage or transportation or they have focused on the techno-economic side. Additionally, review articles on both the LNG and LH2 transportation are not plenty. Lu et al. (2025) conducted a review on key influencing factors on hydrogen storage and transportation costs [46]. Their study extensively focused on technical areas of storage and economic areas of transportation, but did not exclusively include the technical and operational sides of LH2 transportation. Mekonnin et al. (2025) presented an overview of the current state of hydrogen storage methods, and materials. They assessed the potential benefits of and challenges of various storage techniques, but did not focus on transportation [47]. Chilunda et al. (2025) presented an overview of electrochemical cycling of liquid organic hydrogen carriers (LOHC) for hydrogen storage and transportation [48]. They conducted a comparative carbon footprint assessment of electrochemical LOHC cycling processes against thermochemical and hybrid LOHC cycling processes, but did not work with LH2. Shu et al. (2025) conducted an elaborated study combining multiple hydrogen production techniques, efficient hydrogen storage materials, different ways of hydrogen transport and applications of hydrogen, but focus was not given to technical and operational challenges of LH2 transportation [49]. Naseem et al. (2025) presented a detailed review on hydrogen production methods (renewable and non-renewable), hydrogen storage strategies, and advancements in the transportation of hydrogen [50]. Although they included transportation in their study, but it eventually did not include the detailed complex technical and operational issues of LH2 transportation. Yang et al. (2025) conducted a review focusing on hydrogen production, storage, and transportation from renewable energy [51]. They also did not dive into the depth of technical and operational challenges of LH2 transportation. Otsubo (2025) provided a comprehensive review of hydrogen compressor technologies and evaluated the impact of long-distance hydrogen transportation on compressor energy consumption and electrification requirements [52]. He also did not focus the technical side of LH2 transportation. There are also several recent review articles associated with hydrogen storage and transportation [53–60]. All these studies have brought significant impact on hydrogen storage and transportation technologies. However, research on hydrogen energy is still at its early phase. From 2017 to 2023, only 81 papers have been published regarding hydrogen energy. Among those, the groups of operation research, techno-economic-environmental assessments, life cycle assessments, and other groups include 51, 21, 5, and 4 numbers of publications respectively [46]. But, from the authors' point of view, there is no single comprehensive review available which has worked with the detailed thermodynamic and operational challenges and advancements, and integrated cost analysis of ship-borne LH2 transportation.

This review brings the technical, operation, and cost strands together and focuses on ship-borne LH2 transportation. It systematically starts with challenges of ship-borne LH2 transportation, then discusses the core studies and ways of LH2 storage technologies to get insights for transportation, and finally dive depth into the technical and operational discussions of maritime LH2 transportation. It has synthesized current knowledge on LH2 tank thermodynamics, practical ballast and cool down strategies. This review is unique for its comprehensive and technically detailed approach to LH2 transportation which is significantly impactful for researchers and commercials. Conventional reviews primarily summarized information about existing technologies or economic aspects, but have not analyzed from in-depth technical perspectives. Whereas, the core objective of this review is not only

Table-1
Scope and contribution of existing reviews relative to the present study.

Reference No.	Primary scope	Thermodynamic analysis	Maritime operations	Techno-economic Discussions	Key limitation relative to present review
[61]	LH2 transfer operations & safety was the core study area	Qualitative boil-off discussions are conducted	Transfer & venting operations only. Avoiding other critical areas	Not addressed	Focuses on transfer safety. No voyage-scale thermodynamics or economics were discussed
[62]	LH2 storage technologies was analyzed	Boil-off and insulation techniques were reviewed	Transportation losses were noted. But a critical and in depth analysis was not present	Partial. Only tank cost was discussed.	This study lacks maritime operational coupling. Ship-level implications were not considered.
[63]	Insulation structures for LH2 tanks was the main focus	Different thermal protection techniques, especially passive & active systems were analyzed.	Mobile tanks were mentioned	Not addressed	This study treated insulation in isolation, no operational or cost linkage was considered.
[64]	This study compared LNG-LH2 for maritime use	Qualitative thermal requirements were mentioned.	Shipboard handling techniques & materials were included	Qualitative economics was considered	Did not quantify LH2 boil-off evolution. Operational penalties were also not included
[65]	LH2 adoption in maritime transport	High-level energy losses were analyzed	Bunkering, regulations, and safety as operational challenges were discussed	High-level cost barriers were included	Treated boil-off as a static issue. Real-world BOG, and its utilization was not discussed as its significant thermodynamic involvement.
[66]	Entire LH2 supply chain was invlolved	Liquefaction & storage losses were quantified	Transport discussed generically. In depth technical analysis was not considered.	Cost of liquefaction of BOG was emphasized.	Maritime shipping treated as one segment. The study touched almost every sides of the supply chain, so automatically only maritime transportation was not deeply analyzed
[67]	LH2 in transportation It was focused on maritime focus cases.	Qualitative boil-off discussions has been conducted.	Maritime challenges were highlighted	Economic barriers were discussed.	This study lacks thermodynamic-operational coupling, which is necessary to understand real-world scenario.
[60]	Hydrogen in transport. This study focused on road & aviation transportation.	Only storage challenges were outlined	Maritime scenarios were largely absent	Policy-oriented study	Maritime LH2 shipping was not a core focus
[59]	Hydrogen permeation barriers were the core focused area	Material-level mechanisms has been discussed	Hydrogen transportation through pipelines only	Not addressed	This study was not dedicated to cryogenic LH2 shipping systems
[58]	Hydrogen storage for maritime decarbonization was the focused area	Comparative storage options were discussed	Maritime propulsion & ports were outlined	High-level emissions were focused	No quantitative thermodynamic or operational modeling were considered
[68]	This study focused on both hydrogen storage & transportation	Comparative storage & transport techniques were analyzed	Seaborne transport were discussed	Cost comparison across modes was included	This study treated LH2 shipping macroscopically, It did not consider operational thermodynamics
This review	This study is dedicated to LH2 maritime shipping systems only	Coupled boil-off, pressure, heat ingress are extensively discussed	Voyage, heel management, operations are included	Operational cost & energy penalties are explained in depth	This study is the first integrated thermodynamic-operational-economic synthesis considering all the possible factors and areas of cryogenic LH2 maritime transportation

present the state-of-the-art information but also help readers to understand the total technical and operational mechanisms for achieving a holistic knowledge of maritime LH2 transportation technology. Scenarios and thermo-physical analysis of low feel level tank states have received especial focus. This review systematically integrates loading, cool down, voyage, and unloading scenarios which include discussions on BOG generation, operational strategies, and tanks pressurization control, covering all thermodynamic behavior of various ships. It also emphasizes the influence of tank geometry and insulation design which is the basic performance decider for these mentioned performance variables. To understand the novelty of this article more explicitly, a comparative table summarizing overlaps and key gaps of the most recent reviews with key insights would be effective. Table-1 represents the most recent relevant review articles. It includes the primary scope, thermodynamic analysis, maritime operations discussion, techno-economic analysis, and key limitations. This table clearly shows the overlaps and gaps of the most recent reviews relative to the present study.

Along with providing a holistic understanding of maritime LH2 transportation systems, this review offers critical insights into optimizing their efficiency, safety, and economic feasibility. All these inclusions have been based on consolidating the latest research and analytical findings.

2. Challenges of shipborne liquefied hydrogen (LH2) transportation

2.1. Thermal management, tank design and structural constraints, and operational challenges

In the liquid hydrogen (LH2) systems, hydrogen is stored in cryogenic form. One of the most significant challenges in LH2 storage system is the creation of boil-off gas due to heat transfer inside the cryogenic tank from the surroundings. This results in significant amount of hydrogen loss during storage or transportation. The boil-off losses happen due to the heat transfer inside the storage tanks. To minimize it, the necessity for high-performance insulation systems over the whole storage tanks is mandatory [69]. But, even strong insulation is ensured, some heat will always leak in. This causes the evaporation of some part of the stored LH2. To transport hydrogen as liquid form, it needs to be stored at a very low cryogenic temperature (−253 °C) to ensure carrying much more hydrogen compared to hydrogen gas in the same volume. Due to this very low cryogenic temperature, a slight increase of temperature causes the significant evaporation of the liquid hydrogen or the BOG. Using pressurized tanks can be the options to prevent BOG generation as pressurized tanks can keep BOG inside the tanks and have invulnerable behavior against leakage. However, several pressure cylinders are needed to be installed in the ship to store these liquefied energy without BOG ventilation. But this approach brings wasting ship space, increasing the tank weight, operational, and instruments complexity, and capital expenditure (CAPEX). So the storing tanks are designed as lightweight cryogenic tanks to handle very low pressures (low maximum allowable working pressure or MAWP), reduce cost, weight, and ensure excellent insulation [70]. Due to this design constraint, it becomes much more difficult to hold and manage the boil-off gas (BOG) that forms inside the tanks. Again, maintaining extremely low temperatures and proper pressure during all stages, such as storage, loading, and shipping, is difficult in practical cases.

To solve this, ships usually keep a small amount of liquid hydrogen inside the tank after unloading. This amount is called heel which keeps the tanks cold during the empty return trip. It also helps to cool the tanks down before the next loading. But, keeping too much heel reduces the amount of hydrogen delivered. On the other hand, keeping too little heel may not be enough to control tank temperature. So, the core challenge is to find a balance to minimize heat transfer into tanks, reduce boil-off losses, and keep the heel volume as small as possible to transport more hydrogen on each trip [71]. These issues are well-understood for LNG shipping by recent studies. But how these performance indicators would behave or fluctuate in the case of LH2 transportation is still mostly unresolved. Additionally, the common challenges of energy transportation along with the specific challenges for maritime LH2 transportation would need different strategies to be overcome while dealing with LH2 transportation technology. These challenges are originated from the unique properties of hydrogen such as colder storage temperature, different insulation needs, and stricter pressure limits. The possible pathway for solving the challenges of LH2 transportation is still an under-developed area. The optimized heel requirements and the appropriate behavior of heat transfer and stratification compared to LNG are unclear.

To minimize BOG losses, proper balance between storage conditions and operational strategies is mandatory. It becomes a major challenge during long voyages because hydrogen's flammability and cryogenic requirements make design and operation more complex. BOG generation is also affected by the ship's cruising speed and the storage tank's maximum allowable working pressure (MAWP). These parameters also fluctuate during long voyages. To suppress vapor pressure and reduce BOG generation, the increase of the tank pressure is an option, although it imposes limits on allowable fill levels. Again, increasing tank pressure comes with complex structural arrangement and cost requirements. Conversely, to minimize BOG losses, voyage duration can be shortened by higher cruising speed, which again significantly raises fuel demand and operational costs. So, optimizing balance among these parameters is a practical and crucial challenge.

2.2. Optimization of BOG utilization and system integration

The efficient management of generated BOG is another related challenge of liquid hydrogen transportation. Either it can be wasted or collected effectively to use for secondary use. The generated BOG can be used as onboard fuel which would lower the dependence on fossil fuel. But this approach would bring some technical and economic complexities. Engine compatibility to use BOG as fuel is the first challenge for utilizing BOG. Engines should be designed in such a way that it would be run by the generated BOG. After considering engine compatibility, system integration is raised as another challenge. Additionally, it comes with crucial questions including capital cost consideration and complex operational management. So, finding an optimal balance among tank pressure, ship speed, and BOG utilization is a critical design and operational challenge for efficient LH2 transportation.

2.3. Heel management and cool down

Another significant operational challenge for LH2 transportation is the heat ingress for extended periods during the return or ballast voyage after unloading the cargo. After unloading, the tanks become empty and they are exposed to heat ingress which often last for 3 weeks. This causes a gradual rise in tank wall and vapor temperatures. To safely operate the new LH2 receiving stage at the next loading cycle, the tanks must be cooled again below a certain threshold before reloading to very low temperatures [72]. To pre-cool down, a small amount of LH2 is sprayed inside the tank. This lowers the temperature of the tank below the threshold temperature. The process is similar to LNG ships where it takes usually 8-12 h [73–75]. This time depends on tank size and design. This step is performed carefully and made lengthy as rapid cooling may induce thermal stress on tank materials. Sometimes, ships start this pre-cooling process while they are at sea and sailing back to the loading terminal, and don't wait for the arrival to the terminal. But, to perform this, the ships must keep a certain amount of heel after unloading to be used later for cooling. Again, the identification of heel storage amount for cooling requirement becomes a basic challenge. This heel requirement creates a trade-off between operational efficiency and delivery capacity [76]. After return voyage, the temperature difference between tank wall and surroundings also affects the amount of heat transfer into the tank at the beginning of the next trip. But, the temperature of the surroundings fluctuate at different maritime zones and at different weather conditions. So, identifying the temperature difference is also challenging. It is only possible to predict a temperature difference range by considering the intense conditions. So, the determination of onboard cooling requirement always remains a non-specified area, which ultimately affects tank design, heel requirement, and BOG utilization. These operational factors, such as temperature rise during empty voyages, cool down duration, and heel optimization, represent key technical challenges in efficient LH2 transportation.

2.4. Modeling and design uncertainties for large scale LH2 transportation

Beyond operational issues, another critical challenge is the accurate modeling and design of large-scale LH2 tanks. Most existing thermodynamic models for LH2 tanks are simplified and based on lots of assumption, such as uniform temperatures within the liquid and vapor regions. However, when fill levels are low during ballast voyages, both strong vapor stratification and transient heat transfer occur. But those simplified models cannot capture these real-life complex phenomena. This underestimation lead to inaccurate analysis of internal energy, wall temperature, and pressurization rates, particularly when then tanks are mostly empty. Additionally, there is no standardized heat transfer correlation for hydrogen under cryogenic conditions. Different studies consider different tuning parameter, resulting in significant uncertainty of accurate prediction. Moreover, limited experimental validation data for larger tanks act as a driving force for this inaccuracy. Moreover, small Type-C tanks were used in cryogenic liquid carriers (as used in the Suiso Frontier) [77]. But, this type is facing transition towards future large-scale Type-B or membrane-type tanks, introducing new unknowns regarding insulation performance, venting strategy, and thermal stress management [78]. Without reliable modeling frameworks and validated data, optimizing heel mass and cool down strategies remain uncertain. At the same time, the BOG control for next-generation LH2 carriers remains underdeveloped for next generation LH2 carriers.

3. Overview of hydrogen storage technologies

3.1. Hydrogen storage technologies and their challenges

Hydrogen storage technologies vary depending on the operating principles, storage need, energy and infrastructural availability, and the duration needed. So, it is important to present them in a concise manner to better understand the technologies. Inspired from the study of Mekonnin et al. (2025) [47], Table-2 is prepared. It represents the most acceptable, common, and widely studied and used hydrogen storage techniques. It includes storage technologies as compressed gas, liquid state, ammonia, liquid organic hydrogen carriers, and metal hydrides. The table focuses on the major technical advantages and challenges.

Table-2 represents that no single hydrogen storage technology currently satisfies all the required standard criteria. These include high density, low cost, safety, energy efficiency, and scalability. Hydrogen can be stored in all the three states as gaseous, liquid, and solid states to increase energy density. Both above and underground storage facilities are needed for short and long term storage [91]. Figure-1 displays the different hydrogen carrier options for storage, taken from the study of Lu et al. (2025) [46]. In summary, compressed gaseous hydrogen (CGH_2) stored in above-ground is the most common technology for hydrogen storage. Due to their available mature technology of pressure vessels and

Table-2
Hydrogen Storage Technologies and Their Challenges. It also includes the core working principle, operating conditions, and key advantages.

Storage Method	Working Principle	Hydrogen Density (Gravimetric/ Volumetric)	Operating Conditions	Key Advantages	Main Challenges
Compressed Gas (350–700 bar, Type III/IV cylinders) [79–85]	In this method, hydrogen is stored physically as compressed gas. Storage can be done both above-ground and under-ground. To store it for long-time and for large-scale different techniques such as salt caverns, aquifers, depleted oil and gas reservoirs etc. exist [18.9].	~4–6 wt%, ~20–40 g/L	High pressurized hydrogen is stored. Pressure varies from 350 to 700 bars. Temperature remains as ambient temperature	This technique is commercially mature. It offers rapid refueling capacity and relatively simple technology	High pressurized hydrogen needs heavy cylinders to withstand the pressure. It also needs high compression energy demands for pressurizing. Low volumetric density and risk of embrittlement are also the challenges.
Liquid Hydrogen (LH2) [86–89]	Hydrogen is stored as liquid state at cryogenic temperatures (~20 K)	~7–8 wt%, ~70 g/L	Due to the cryogenic temperature, advanced insulation is required to resist any heat ingress inside the storage tank.	Liquid hydrogen offers high volumetric density. It is also favorable for large-scale transport	The liquefaction process of hydrogen consumes around 30 to 40% of stored hydrogen energy. Boil-off losses, stratification, complex insulation etc. are the major challenges. Challenges are described elaborately in Section 2.
Ammonia (NH_3) [86–89]	In this technique hydrogen is not stored as unique compound, rather it is chemically bound in ammonia (17.6 wt% H_2)	108 kg H_2/m^3 equivalent	The storage pressure is not too high like CGH_2. The compound is stored with mild pressure of around 10 bars or refrigerated liquid	Ammonia has high volumetric density. It offers zero-carbon carrier and are acceptable with existing global infrastructure	Toxicity is the most significant challenge. The cracking is also energy-intensive. Additionally, while decomposing H_2, impurities are traced in most of the cases.
Liquid Organic Hydrogen Carriers (LOHCs) [86, 87,90]	In this method, liquid Hydrogen is stored and released by reversible hydrogenation–dehydrogenation technique. This storage system uses reusable catalyst for hydrogenation, and a catalyst for dehydrogenation.	5–7 wt%	Ambient liquid state, catalyst-assisted reactions	This system offers high safety. There is no issue of boil-off gas generation. It is also compatible with existing fuel logistics.	Dehydrogenation needs high energy input. The catalysts are degraded over time and have limited cycle life.
Metal Hydrides (e.g., MgH_2, $LaNi_5H_6$) [86, 87]	In this process, hydrogen is absorbed into solid crystalline lattices	1–7 wt%	Temperature is slightly higher, around 200 to 400 °C. Pressure also remains moderate.	This system offers high volumetric density. It is also compact and safe.	This system uses heavy materials. It has slow kinetics and difficulties of heat management and cost

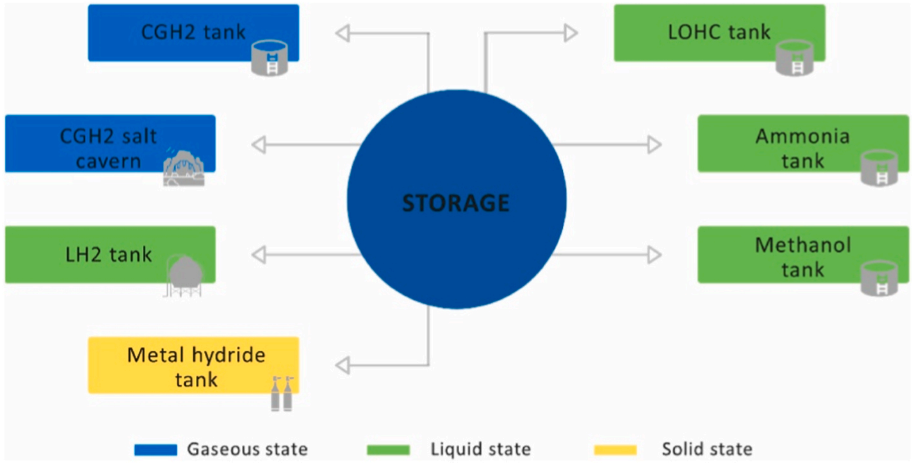

Figure-1. Schematic diagram of most common and acceptable hydrogen storage technologies. It can be stored in all three forms of matter: solid, liquid, and gaseous states. The active states or phases are shown three colors. Blue, green, and yellow colors represent gaseous state, liquid state, and solid state respectively [46].

commercial availability have made this technique widely acceptable than other storage technologies [79–85]. CGH_2 tanks are designed to carry high pressurized CGH_2, but still the Low storage capacity is the main challenge of CGH_2 tank options. Storing as salt caverns at underground is one of the most potential storage technologies due to their technological and commercial availability. This technology has also been researched plenty of times [46].

3.2. *Foundational studies on the thermodynamic and operational behavior of LH2 storage tanks*

It is essential to first examine the technical and operational characteristics of LH2 storage for the better understanding of LH2 transportation technology, as transportation occurs through storing first. Many of the fundamental processes including thermal stratification, BOG formation, pressurization, and chill-down phenomena follow the same principles for both LH2 storage and transportation. For this reason, Table-3 is added presenting the fundamental studies of LH2 storage with their focused area and key finding, although this review is focused on LH2 transportation. The table has covered some of the key studies conducted in the first couple of years of foundational research of LH2 storage technologies. The studies have covered LH2 tank thermodynamics, filling operations, and heat transfer mechanisms to better understand the thermal behavior, phase transitions, and pressure evaluation. These all are crucial for designing and operating efficient cryogenic storage systems in LH2 carriers.

4. Comparison of alternative hydrogen transportation technologies

To analyze why the researchers are opting more for LH2 transportation studies recently, the other transportation technologies need to be studied. Table-4 represents the four major hydrogen transportation technologies other than LH2 transportation. It includes transportation through pipelines, NH_3 shipping, liquid organic hydrogen carrier (LOHC) shipping, and cryo-compressed hydrogen. The table explains description, efficiency or losses, infrastructural needs, and limitations of each transportation techniques.

From Table-4 it can be said that hydrogen transportation models are highly context dependent and have no universal optimal solution. Pipelines offer efficiency for regional transport but become less-feasible for long distance transport (over 2000 km). This technology also requires costly material upgrades. Ammonia and LOHC can utilize existing infrastructure but their application is limited due to the challenges of cracking and dehydrogenation. Cryo-compressed technology is an emerging method but brings complex tank design and operational requirement for very low temperature and ultra-high pressure. Although LH2 transportation has its own challenges and limitations too, it has emerged one of the most acceptable technologies through overcoming most of the challenges of these summarized technologies.

5. Thermodynamic understanding of heel management: from LNG carriers to LH2 systems

There are several studies regarding heel management of LNG carriers in ballast voyage. Studies suggest that, in 2018, most LNG carriers were membrane-type covering 70% of the total LNG carriers. Hasan et al. (2009) demonstrated that the quantity of LNG required to cool a membrane-type tank from ambient temperatures may range from 800 to 1600 m^3. It is also estimated as 0.5-1% of total carrying capacity. They concluded that it is common practice to carry a heel of around 5% [105]. Krikkish (2018) carried out theoretical and experimental analysis where he studied the nature of pressure fluctuations inside a LNG tank. He showed that due to uneven tank wall heat fluxes the pressure fluctuation happens. He also stated that the liquid temperatures remained stable while gas temperatures varied. He concluded that the system tends towards thermodynamic equilibrium in ballast voyages [106]. Krikkis et al. (2021) [107]conducted a study on how LNG behaves inside ship tanks during the laden voyage (when the ship is full of cargo) and the ballast voyage (when the ship is empty after unloading, but still carries a small amount of liquid to keep tanks cold). They developed a combined heat transfer and thermodynamic model to understand how the insulation of the tanks interacts and impacts the thermodynamic properties of the LNG cargo. They highlighted several technical challenges in LNG shipping. Many of these challenges are also relevant for future hydrogen carriers. These include: multiple boiling regimes occurring at the same time on the tank surface due to the non-uniform heat gain on different parts of the tank wall; the difficulty to predict sudden pressure changes inside the tanks during laden voyage which are linked to turbulent vapor release from non-uniform boiling on the insulation surface; the difficulty of predicting cargo behavior during laden voyages due to multi-phase and multi-boiling behavior. They commented that the loading and discharging cycle is crucial for understanding how tanks behave across both voyages. During ballast voyage, the cargo is much closer to thermodynamic equilibrium. They also stated that the liquid temperature stays almost constant but the gas temperature varies significantly. In some tanks, they observed small-scale rollover phenomenon due to this [106–108]. Additionally, the boil-off rate is not fixed per day. Rather, it is directly linked to the ship's propulsion system and speed-power curve. They concluded that a single tank usually carries the main heel, while

Table-3
Summary of key studies on thermodynamic behavior and operational challenges of LH2 storage systems in its development phase.

Author(s)	Study Focus	Key Issues Addressed	Key Findings
Molkov, Dadashzadeh, Makarov (2019) [92]	This study focused on the physical model of onboard hydrogen storage tank thermal behavior during fuelling	They addressed temperature rise during fast fuelling, and pressure buildup inside tank. Other challenges they specified include short fuelling times, complex heat transfer, and limitations of previous models	Their model predicted temperature fluctuation within ±5 °C. It incorporated real gas behavior, and entrainment theory for velocity. It was suitable for automated fuelling protocols
Liu & Li (2019) [93]	Core objective was to analyze thermal physical performance in LH2 tank under constant wall temperature	They addressed thermal stratification, and phase change effects. Other key issues the considered include rapid pressurization, influence of gravity, wall temperature, and liquid height	Their results demonstrated that phase change significantly affects pressure. Additionally, gravity accelerates stratification, and wall temperature impacts stratification rate. They commented that liquid height affects development
Petitpas (2018) [94]	They worked with BOG losses during LH2 transfer at refueling stations	Intrinsic losses from heat ingress were the main concern of their study. They also considered complex thermodynamics, unclear loss mechanisms, and cost impact	Results revealed that losses mainly occurred from vapor compression. These losses can exceed 12%. They commented that liquid density remained non-ideal.
Zuo, Sun, Jiang, Qin, Li, Huang (2019) [95]	They analyzed thermal stratification suppression in LH2 tanks. They considered self-spinning spray bar for this suppression	Their core concerns were thermal stratification, and inefficient mixing. They also specified payload/power constraints in spacecraft	Study demonstrated that self-spinning spray bar reduces temperature differences, and causes 70% payload savings.
Liu, Li, Zhou (2018) [96]	They focused on thermal stratification in LH2 tanks under different gravity levels	They specified challenges such as strong stratification in low gravity, localized evaporation, and surface tension effects which are dominant at microgravity.	Results exhibited that gravity reduces stratification in 1g, and microgravity increases layering. They CFD based results were validated with AS-203 flight data. They concluded that surface tension effects are significant.
Melideo, Baraldi, De Miguel Echevarria, Acosta Iborra (2019) [97]	They studied the effect of injector diameter, direction, and gas temperature on thermal stratification during filling	Their concern include: high initial gas temperature increases stratification, downward injectors worsen stratification, and large diameters reduce mixing	Results showed that upward injectors reduced stratification, and smaller diameter promoted mixing. Additionally, critical velocity thresholds depend on injector direction. Wall temperatures were also affected with this.
Ma, Zhu, Li, Xie (2020) [98]	They studied the phenomena of chill-down and thermal stress in LH2 tank during ground filling	Key focused challenges include; rapid wall cooling causes large thermal stress, filling rate affects stress distribution, and risk of structural failure	Their CFD plus structural analysis predicted temperature gradients and stress. They revealed that bottom was critical at start, and middle was critical later. They also commented that filling rate correlates with stress
Zhou, Zhu, Hu, Wang, Xie, Zhang (2019) [99]	They studied the effect of ullage pressure on LH2 evaporation rate	They specified that heat ingress causes pressurization, and safety valve pressure influences evaporation. They also considered the complex internal flow patterns.	Results demonstrated that evaporation occurs in 3 stages, and rated ullage pressure mainly affects liquid-invariant period.
Liu, Feng, Lei, Li (2018) [100]	Their objective was to investigate thermal stratification under sloshing excitation in non-isothermal LH2 tank	They addressed the disturbance of stratification by sloshing. They also considered heat leaks and phase change impact and safety risks of LH2 tanks.	They found that sloshing mixes layers which affects pressure and temperature distribution. Their results provided design insights for sloshing suppression.

the remaining tanks are kept close to or entirely empty. And, vapor-only tanks may reach ambient temperature over the continuation of the voyage, but the heel carrying tank will remain significantly colder [107]. Overall, this study suggested that LNG ship transportation cannot be understood by studying the laden and ballast voyages separately only, rather they are connected by the loading and discharging cycle. Although, the main challenges are based on managing heat transfer and pressure fluctuations during laden voyage, the ballast voyage is easier to model due to its equilibrium-close behavior. The study also emphasized that boil-off rates depend on not only cargo properties but also on ship operations. Kulista and Wood (2020) marked intermittent spraying in the 3 days prior to arrival for loading as a common practice [109]. These understandings are very important sight for future hydrogen transport studies regarding heel management.

6. Effect of tank geometry on the thermodynamic performance of LH2 storage systems

To understand the thermal performance of liquid hydrogen storage tanks, Wang and Merida (2024) [110] conducted an analytical study where they compared different tank shapes (spherical, vertical cylindrical, and horizontal cylindrical). They focused to analyze the impacts of geometry on pressure rise, boil-off gas (BOG) generation, and overall thermal behavior. They developed a non-equilibrium thermodynamic

Table-4
Summary of major transportation options and limitations of hydrogen.

Transportation Method	Description	Efficiency/Losses	Infrastructure Needs	Limitations
Pipelines [101]	In this technique, hydrogen is transported under pressure through natural gas pipelines. These pipelines are dedicated or repurposed for transportation only.	Transport through pipelines offer high efficiency for regional transport.	Pipelines demand some basic engineering pre-requisites. They need hydrogen-compatible materials to avoid the risk of embrittlement, and compressors to create sufficient pressure difference for smooth flow.	Ley limitations include: risk of embrittlement and leakage. Long-distance pipelines also need very high capital investment.
NH_3 Shipping [102]	Hydrogen is transported as ammonia. Later, it is cracked at destination.	This technique is efficient for bulk transport. However, during unloading or extracting hydrogen from ammonia, significant energy is required for cracking	This process needs mature global ammonia infrastructure.	Key drawback is the toxicity hazards of NH_3. Additionally, cracking offers significant cost, and purity concerns.
Liquid Organic Hydrogen Carrier (LOHC) Shipping [103]	In this technique, hydrogen is transported through liquid carriers and transported like oil. It relies heavily on catalysts, which are used for hydrogenation (storage) and dehydrogenation (release)	This method offers no boil-off losses. It also provides reversible reaction cycle.	LOHC shipping is compatible with fuel transport infrastructure.	The most significant drawback is the high dehydrogenation energy input. Additionally, catalyst degradation and limited cycle life are major constraints.
Cryo-compressed Hydrogen [104]	This process needs hybrid of cryogenic storage. Hydrogen is stored and transported through highly pressurization.	This cryo-compressed hydrogen has higher density than either compressed or liquid hydrogen alone. This helps to carry more amount in a small space	This system needs advanced tank technology, as the combination of pressurization and cryogenic temperature brings several design and operational challenges.	This technology is still experimental, and complete infrastructure has not been established yet.

model and did not assume that liquid vapor phases are always at the same temperature. They found that spherical tanks performed best because they had the lowest pressure increase rate in self-pressurized conditions. It happened due to their smallest surface area for the same volume which reduced heat transfer from the environment. As a result, spherical tanks showed the lowest BOG rate. Horizontal cylindrical tanks performed better than vertical ones as horizontal tank-type offers slow pressure rise than vertical cylinders. This occurred due to more effective heat transfer between vapor and liquid phases inside the tank. They found that higher fill levels help to suppress evaporation and pressure rise because a higher initial liquid level increases the thermal mass of the liquid. So, it takes more heat to raise the temperature and cause evaporation. Additionally, the BOG rate was lowest in spherical tanks at atmospheric pressure. But, both horizontal- and vertical-cylindrical tanks showed almost the same BOG rate under atmospheric condition.

Although their study offered valuable insights regarding the effects of LH2 tank size on the thermal performance of the tanks, some unresolved challenges exist. Spherical tanks showed better thermodynamic performance, but such tanks with good geometry still faced continuous heat ingress, which indicates the insulation limitations for perfect elimination of BOG. Their study mostly considered stationary storage or simplified thermal loads. But, moving ships experience different practical phenomena, such as sloshing, vibrations, varying external conditions, which were not considered in their study. Moreover, they addressed BOG management, but the risks of hydrogen leaks and explosions in real-world environments were not studied. These unresolved challenges open the pathway for potential future research directions including BOG suppression strategies, large-scale validation, dynamic behavior studies, optimization of advance insulation materials and tank geometry etc.

7. Comparative analysis of liquid energy carriers and boil-off gas (BOG) losses

Al-Breiki and Bicer (2020) compared five potential energy carriers that can transport energy (especially from natural gas) over long distances in liquid form [111]. The five carriers include Liquefied Natural Gas (LNG), Dimethyl Ether (DME), Methanol, Ammonia, and LH2. The study focused on how much BOG each carrier loses during the supply chain (land storage → loading → ocean transport → unloading). Additionally, to analyze the factors which affect the losses was also a major objective of that study. They performed a technical analysis and sensitivity study to calculate daily BOG rates for each carrier, evaluate BOG generation at different stages of transport, see how external factors such as temperature, pressure, voyage time, heel percentage, and pumping time affect BOG formation, and compare how much energy each carrier can effectively deliver after accounting for those losses. Figure-2 shows the total supply chain of hydrogen transportation and the challenge of BOG generation in every phase. Heat ingress happens in every phase which causes significant amount of hydrogen losses through boil-off [111].

They found that hydrogen showed highest daily BOG losses as around 3.438%, and methanol showed lowest daily BOG losses as around 0.049%. LNG demonstrated the second highest daily BOG losses as 0.471%, but it is still far less than for the case of hydrogen. Among the different stages of the supply chain, ocean transportation phase offered the main source of losses due to the long exposure time and large tank surface area. They concluded that DME carried the most energy efficiently, Methanol had the least storage losses, and Hydrogen had the worst performance in terms of mass loss due to evaporation [111].

These findings are based on large-scale, long-distance marine transportation under fully refrigerated conditions. All the energy carriers are assumed to be transported at atmospheric pressure (≈1 bar) in cryogenic tanks. They assumed a fixed ship storage capacity of 160,000 m^3. It consists of four spherical tanks with a radius of 21.23 m. These assumptions enable a fair comparison across carriers. For LH2, this corresponds to a total stored mass of approximately 2.84×10^6 kg. The baseline voyage duration is assumed to be 30 days. This duration represents long-haul overseas routes and identified as the dominant contributor to total BOG generation. Passive insulation is considered, with an overall heat transfer coefficient of approximately 0.1 W/m^2K for hydrogen tanks. In the reference case, no active BOG re-liquefaction or utilization is assumed. Under these conditions, the extremely low boiling temperature of hydrogen results in a large temperature gradient between the stored medium and the ambient environment. The study assumed ambient temperature variations between 10 °C and 50 °C, and considered an increase of BOG rates by approximately 0.1%. But, increasing storage pressure from 1 bar to 70 bar, reduces total BOG losses by about 0.3%. Additionally, higher heel percentages and longer

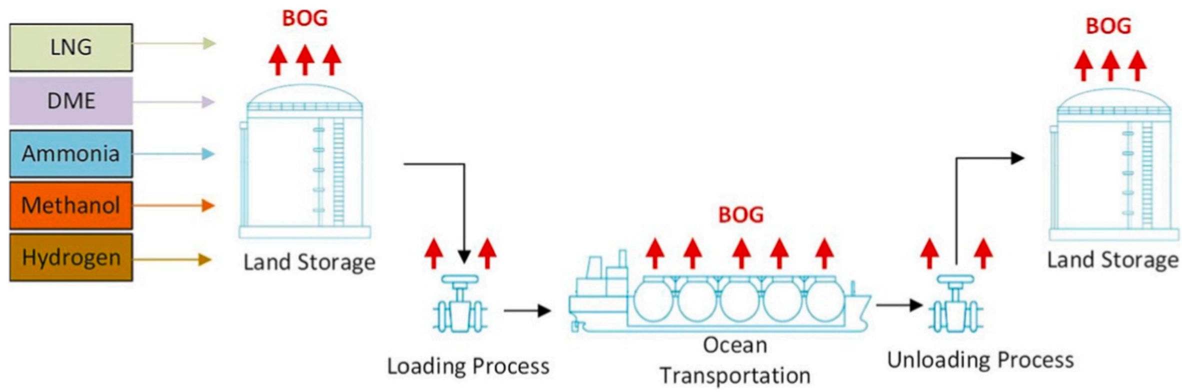

Figure-2. Schematic diagram of supply chain of Hydrogen transportation system. The supply chain starts from the land storage of hydrogen, and then continues to loading process, ocean transportation, unloading process, and finishes with land storage again. Among these phases ocean transportation offers the most significant BOG generation due to long term exposure to environment and large tank surface area [111].

pumping times during loading and unloading phases further increases LH2 boil-off. This is occurred due to the increased thermal interaction and auto-refrigeration effects. The absolute magnitude of the daily BOG rate is sensitive to tank design, insulation performance, storage pressure, voyage duration, and BOG management strategies. But, hydrogen consistently exhibited the highest boil-off losses among the evaluated carriers across all examined scenarios [111].

8. TECHNO-ECONOMIC analysis of LH2 transportation and boil-off gas (BOG) utilization strategies

8.1. *Comparative cost assessment of liquid energy carriers*

Transporting LH2 is expensive and technically difficult. The economic feasibility of large-scale LH2 transport comparing to other natural-gas-derived energy carriers shows significant challenges. To reduce the BOG generation and the transportation cost, several options, such as storing LH2 under higher pressure to decrease venting, or using BOG as fuel for the ship instead of venting, or sailing the ship faster to shorten voyage time, can be adopted. But all these choices affect economic (NPV) and cargo delivery. To make a comparative cost assessment of the transportation of different natural-gas-derived energy carriers, Al-Breiki and Bicer (2020) conducted another analytical study, considering both BOG losses and the social carbon cost (SCC) [112]. They analyzed five liquefied carriers: LNG, DME, methanol, ammonia, and LH2. Figure-3 demonstrates the graphical abstract of their study which represents the overall cost structure. It is segmented into two sections: the upper section showing the cost of the conversion of natural gas into liquefied forms, and the lower section showing the transportation costs to demanded regions. All the costs are classified into Capital Cost, Operations Cost, Boil-Off Gas Cost, and Social Carbon Cost, where the Operation Cost for transportation includes various charges (e.

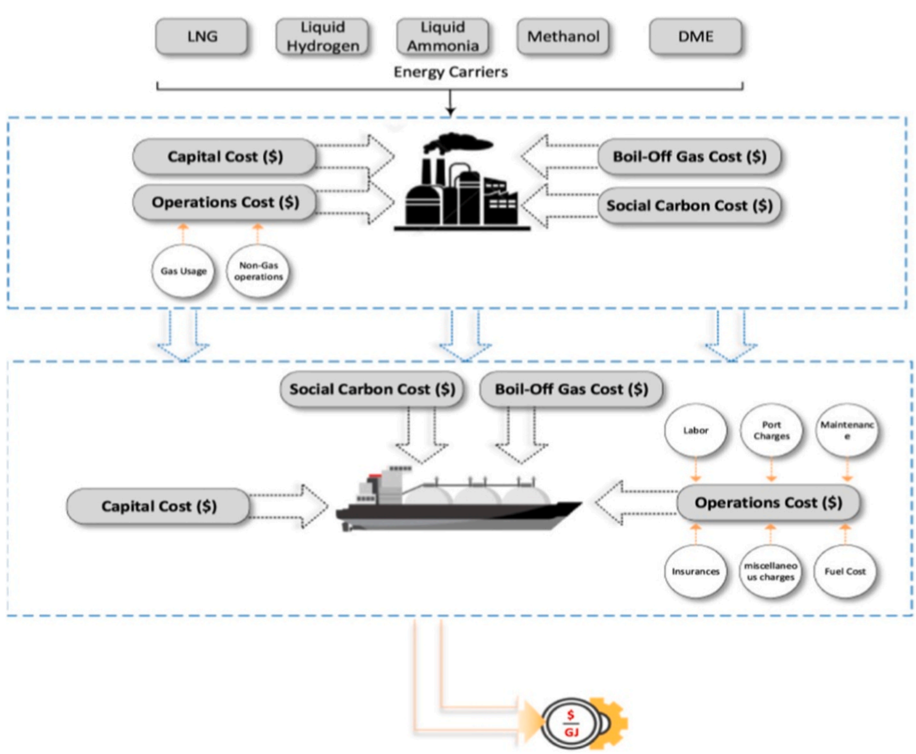

Figure-3. Graphical abstract of the study conducted by Al-Breiki and Bicer (2020) [112]. It is segmented into two critical cost-phases of the supply chain: Production Cost of converting natural gas energy into a liquefied form of energy (upper section), and Transportation Cost (lower section). The cost analysis includes five energy carriers: LNG, LH2, liquid ammonia, methanol, and DME. All are derived from natural gas.

g., port charges, fuel cost, insurances etc.). The generated BOG during production of a liquefied energy carrier is managed by two ways; either it is vented to the environment or sent to a re-liquefaction facility. In this study, this lost BOG is accounted for within the total production cost. The same approach is applied in the case of transportation where the lost mass due to BOG is accounted as a cost. The figure is added to represent the holistic cost assessment instead of just showing simple costs regarding fuel prices or shipping fares. The most significant feature of this figure is the addition of BOG cost and Social Carbon Cost (SCC) which is the two major challenging factors for liquid energy transportation, especially for LH2 transportation.

Figure-4 shows the results of the cost analysis of the mentioned five different energy carriers in \$/GJ. In every case, the costs are divided into Capital Costs, Operating Costs, and BOG Costs. It demonstrates that LH2 exhibited the highest transportation cost of about 3.74 \$/GJ, even after accounting for carbon externalities. This amount is substantially higher than LNG (0.74 \$/GJ) or liquid ammonia (1.09 \$/GJ). Among these five cases, the cost of LH2 in all the three departments (Capital Costs, Operating Costs, and BOG Costs) is the highest. Several factors, such as the low volumetric energy density, extreme cryogenic temperature requirement (−253 °C), and significant BOG losses during long voyages due to extremely sensitive to heat, are primarily responsible for this high transport cost of LH2. The study also indicated that the inclusion of SSC favors carbon-free carriers such as hydrogen and ammonia. But the economic advantage of LH2 does not show sufficient potential, as the BOG-related energy and mass losses dominate the cost structure [112].

The cost figures are derived under some clear baseline assumptions. For the comparative carrier assessment, the assumptions are: base natural gas price is 2 \$/GJ, ship capacity is 160,000 m^3, fixed cruising speed is 20 knots, and a representative long-haul distance is 12,000 km. The sea-route is assumed from Qatar to Japan [105]. Liquefaction, storage, and transportation costs explicitely account for BOG generation. This is true for both production and shipping phases and is treated as an economic loss rather than an externality. Fuel prices for marine operations are based on heavy fuel oil at 0.58 \$/kg. At the same time, the environmental externalities are incorporated through a social cost of carbon (SSC) baseline of 46 \$/t CO_2eq and consistent with widely adopted mid-range estimates [112].

But, variations exist in key uncertain parameters. These include natural gas price (1-4 \$/GJ), shipping distance (2400–12,000 km), ship capacity (100,000–250,000 m^3), and social carbon cost (13-137 \$/t CO_2eq). Across these ranges, LH2 transportation consistently remains the most cost-intensive option on a per-unit-energy basis. This is especially true for long-haul routes. This is primarily due to its Low volumetric energy density, high liquefaction energy demand, and significant BOG losses of LH2 are the primary reasons for this most intensive cost requirement. But, in the case of short-distance routes, low hydrogen production costs, or high carbon pricing regimes, the scenario changes much. In that cases, LH2 and liquid ammonia become increasingly competitive relative to LNG. These results indicate that absolute cost values are sensitive to future market and policy conditions, but the qualitative ranking of energy carriers and the dominant cost drivers (liquefaction energy demand, BOG management, carbon pricing) would remain robust across potential future scenarios. And these robust conclusions are not newly assumed in this work, rather they are evaluated from the original study of Al-Breiki and Bicer (2020) [112].

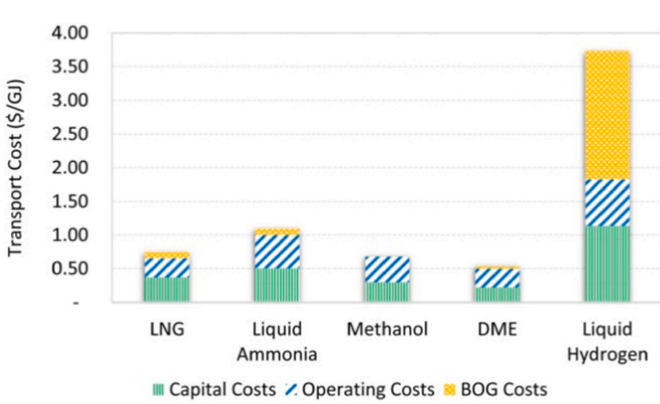

Figure-4. Graphical representation of the transport cost (\$/GJ) of energy transportation. It includes the transportation cost of LNG, liquid ammonia, methanol, DME, and liquid hydrogen. The costs include three distinct types of costs: capital cost (represented by green color), operating cost (represented by white color with blue lines), and BOG cost (represented by yellow color) [112].

8.2. THERMODYNAMIC-ECONOMIC coupling for BOG utilization

To analyze the cost constraints and the most effective option to handle BOG, Wang et al. (2025) conducted a study using thermodynamic storage model and ship fuel consumption model to simulate voyages [113]. They analyzed the feasibility to recover and use of BOG instead of being vented and wasted. They examined the potential to use this BOG for ship propulsion or onboard power generation as a secondary energy source to enhance voyage efficiency and reduce fuel cost. Their study looked at a 160,000 m^3 LH2 carrier. Their thermodynamic and economic model combined three sub-models which are presented by Figure-5. It includes a storage BOG model to estimate BOG generation rates, a propulsion model based on ship cruising power, and a ship resistance model to determine fuel consumption [113].

Figure-6 demonstrates the overall powertrain of the ship, which is a combined gas and steam (COGAS) turbine system that generates electricity [113]. The engine powers a generator and then the generator powers an electric motor that turns the propeller. The figure highlights that power is needed for both propulsion (P_{thrust}) and for auxiliary systems($P_{auxiliary}$). The powertrain includes a fuel cell which is an optional component to provide the auxiliary power. It potentially uses BOG directly with high efficiency. In the figure, $P_{aux,fc}$ is the electrical power for the ship's auxiliary purpose, supplied by the fuel cell. $P_{aux,COGAS}$ is the auxiliary power supplied by the COGAS engine's generator. $P_{brake,COGAS}$ is the total mechanical power output from the COGAS engine's rotating shaft, which is split between propulsion and auxiliary needs. $P_{thrust,COGAS}$ is the portion of the COGAS engine's electrical output that is dedicated to the propulsion motor to drive the ship forward. $P_{eff,thrust}$ is the final useful power after accounting all losses. It actually pushes the ship through the water.

They proposed four operational modes to simulate how BOG can be utilized during voyages which are represented by Figure-7. Mode A offered conventional operation using marine gas oil (MGO). In this case, all BOG was vented and burned in a gas combustion unit. Mode B was similar to Mode A but included a proton exchange membrane (PEM) fuel cell, which was powered by BOG for auxiliary power generation. Mode C was based on full BOG-fuelled propulsion. In this case, all generated BOG fed the combined gas and steam (COGAS) engine. Forced BOG generation was applied when the BOG rate was lower than the propulsion demand. This was done by vaporizing additional LH2. Mode D offered hybrid approach. It used BOG for propulsion which was supplemented by an onboard LH2 fuel tank when natural BOG generation was insufficient, especially during ballast voyages [113].

Figure-6 and Figure-7 are added to understand the possible pathway to use BOG while transporting LH2 instead of wasting. They demonstrate that BOG is a fuel, not just waste. They also exhibit that system integration is the key which indicates the choice of propulsion system (COGAS, fuel cell) is deeply interrelated with cargo tank management. Figure-7 also shows the operational flexibility by depicting different modes of BOG uses based on the fluctuation of BOG generation.

They also investigated how tank pressure (MAWP), and cruising speed affect BOG losses, fuel costs, and net present value (NPV). LH2 transportation offers complex pathway and inherent losses at every

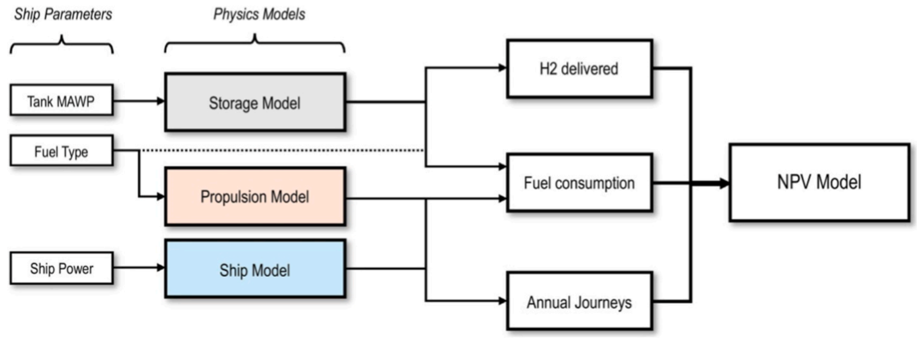

Figure-5. Thermodynamic and economic model including three sub-models: storage model, propulsion model, and ship model. It shows the relationship between ship parameters, the ship and its resistance, propulsion, cryogenic storage models, and broader economic model. The authors of this model stated that the dependence of fuel consumption estimation on the storage model output was applied only when hydrogen was used as a fuel [113].

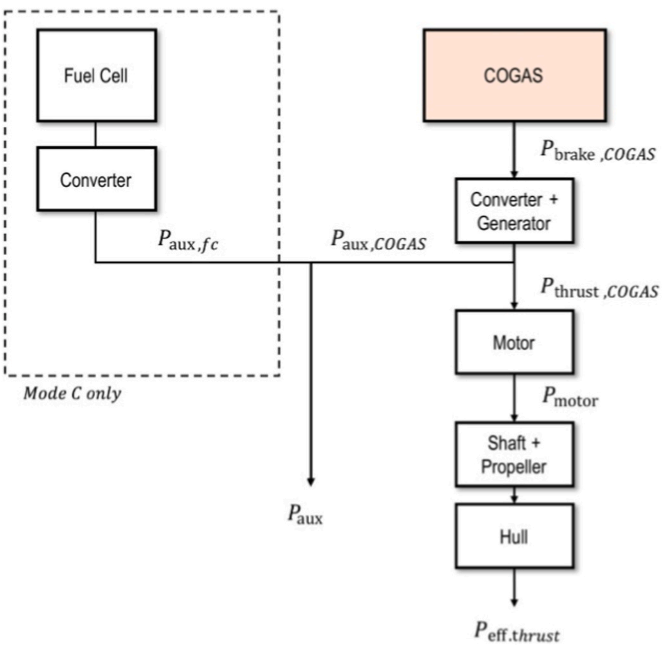

Figure-6. Integrated powertrain architecture for a COGAS-electric LH2 carrier. It shows the general power flow from fuel to propeller. It integrates both the main COGAS engine and a potential fuel cell for auxiliary power [113].

stage. To analysis the losses Figure-8 is added which maps the hydrogen streams and expected losses [113]. The losses include both liquid and gaseous phase across a single round-trip journey. It starts from production and liquefaction to the laden voyage, unloading, and the return ballast voyage.

The comparison between conventional marine gas oil (MGO) and hydrogen was also done to understand how they affect the mentioned terms. Their study focused on some challenges; higher tank pressure reduces BOG but lowers allowable fill level, higher speed reduces losses but increases fuel costs, and uncertainty in carbon pricing and hydrogen market price. They built a physics-based analytical model to simulate tank heat gain and BOG with different MAWPs, and to analyze fuel use at different ship speeds. They also conducted some case studies for various voyage distances, hydrogen prices, and fuel prices. Their study demonstrated that higher MAWP improves storage efficiency but lowers fill level. Ship speed didn't affect much to BOG reduction. The main effect of ship speed is more annual deliveries. In the case of fuel choice, MGO is cheaper when carbon price is not considered. But hydrogen fuel becomes competitive when carbon price is considered. Another key finding was when ship needs more fuel than natural boil-off provides, forced boil-off is required. But this technique showed less efficiency. For short voyages, there's no big need to store hydrogen under high pressure. And, when hydrogen pricing is high, it's better to keep as much liquid hydrogen as possible rather than lose it as BOG. In both cases, operating the storage tanks at atmospheric pressure is more economical than operating at increased pressure. Additionally, optimizing both ship speed and MAWP could add up to 180 million USD net present values (NPV). They concluded that hydrogen production cost, and voyage distance are the main drivers for economic benefits, not ship parameters [113].

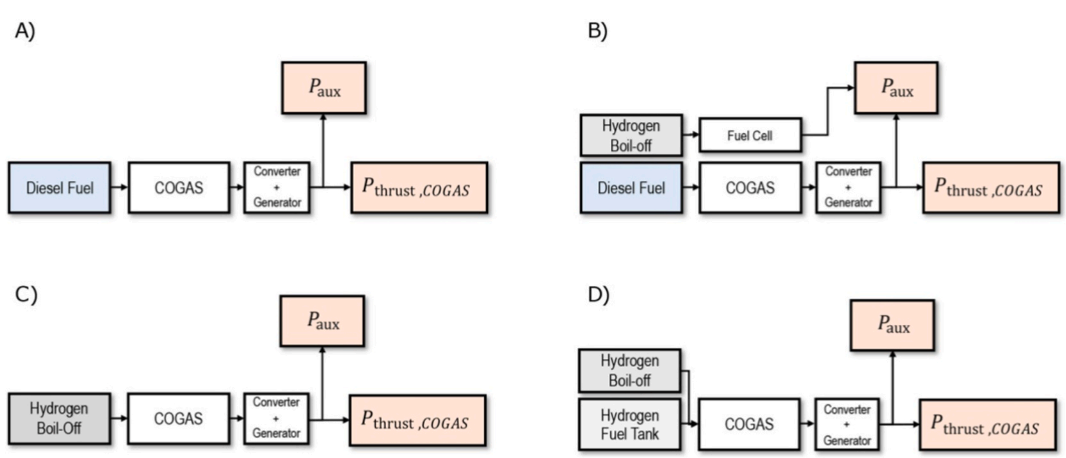

Figure-7. Operational modes for BOG utilization in LH2 carrier propulsion. It comprises schematic of the four fuel management strategies: (a) conventional MGO, (b) COGAS engine's power with BOG for auxiliary power via fuel cell, (c) BOG for main propulsion, and (d) BOG with an LH2 fuel tank to cover fuel deficits [113].

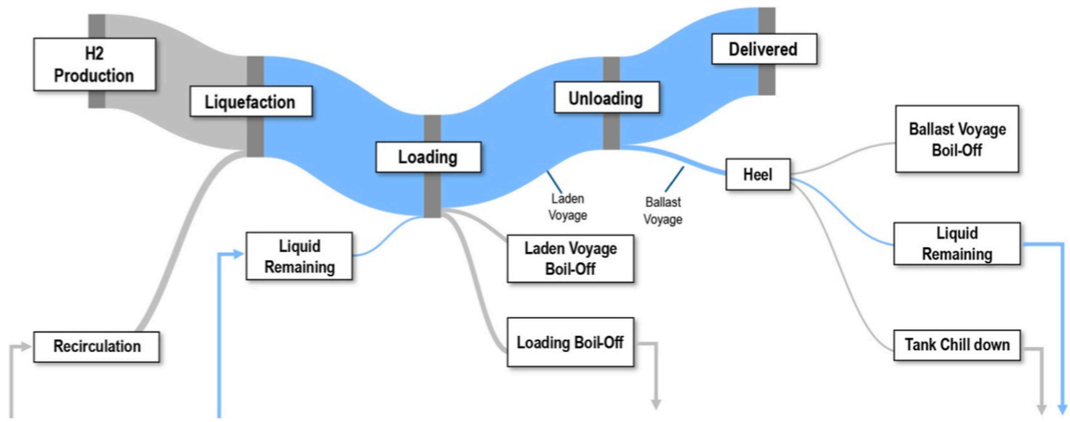

Figure-8. Schematic diagram of the LH2 export pathway. It was analyzed by Wang et al. [113] and they focused on the loss mechanism. The diagram presents the continuous BOG formation during storage (liquefaction, heel) and voyages, and the discrete losses during different operational phases such as loading, unloading, and tank chill down The grey and blue colors represent gaseous and liquid streams, respectively.

This study was the first one to analyze the interaction between tank MAWP, fuel type, and speed. Previous studied didn't combine all these three variables. But this study offers some unresolved challenges. The data on LH2 ship construction cost (CAPEX), and how MAWP and insulation affect it were limited. Several variables including hydrogen selling price, carbon tax, and fuel costs fluctuate a lot which show uncertainty in hydrogen markets. Hydrogen engines and onboard BOG management are not still mature which paves the way to conduct more studies on hydrogen engines.

8.3. Scenario-based assessment of carbon pricing and route length

Wang et al. (2025) explicitly evaluated fluctuating carbon prices in their study. They also included multiple shipping routes in their analysis. In their study they examined a range of carbon price scenarios, which was approximately 70 to 250 \$/t-CO_2eq, through their impact on effective marine fuel cost [106]. The results demonstrate a clear understanding that without carbon pricing conventional marine gas oil (MGO) remains economically preferable. But, under moderate to high carbon prices, such as around 70-150 \$/t-CO_2eq and above, hydrogen-fuelled propulsion becomes increasingly favorable. This yields NPV gains of up to 180 million USD under high-carbon-price scenarios [113].

Similarly, they assessed route-dependent scenarios by varying voyage distances between 4000 and 12,000 km [113]. These assessments correspond to representative short-, medium-, and long-haul shipping routes. Shorter routes reduce cumulative boil-off losses and favor lower tank pressurization. On the other hand, long-haul routes amplify the importance of cruising speed optimization and boil-off management. Although absolute cost and NPV values vary across these scenarios, the dominant economic drivers remain constant. These include voyage distance, hydrogen production cost, boil-off gas utilization strategy, and carbon pricing. All these assessments indicate that the qualitative techno-economic conclusions for LH2 shipping are robust across a wide range of long-term operation al scenarios. This is also true for LH2 based policy scenarios.

The analyses presented in Section 5-8 are based primarily on theoretical models, numerical simulations, and extrapolation from LNG operational experience. These models provide insights into heel management, tank geometry, BOG generation, and economic feasibility of LH2 shipping. But, they are not directly validated against full-scale operational LH2 carriers. The current limited availability of full-scale vessels have made it difficult to real-world experimental validation. Therefore, the results should be interpreted as predictive and indicative, rather that fully verified at commercial or pilot-ship scale.

8.4. Operational challenges and economic implications of LH2 shipping for maritime operators

The discussed theoretical and simulation-based studies provide valuable insights on BOG losses and transportation costs for LH2. But shipping companies face real-world operational challenges which can significantly affect performance and economics. Ferrari et al. (2023) showed in their study that there is a rise in maritime transport costs during the 2021-2022 global shipping surge. They showed it to demonstrate the increasing shipping rates by up to 30-40% containerized and bulk cargo. This led to reduced trade flows and increased commodity prices [114]. Regions with limited port infrastructure and inflexible logistics experienced disproportionate impacts. These highlight that operational constraints, such as port congestion, limited berth availability, and reduced options for alternative shipping routes, can amplify the economic burden of transporting sensitive energy carriers like LH2.

In addition, shipping companies' strategic decisions under cost pressures can further influence operational outcomes. Wu et al. (2025) analyzed the digitalization of the shipping industry. They demonstrated that investment costs and operational efficiency interact to produce nonlinear effects on freight rates and profitability [115].For example, unilateral adoption of efficiency-enhancing digital technologies can yield a 'winner-takes-all' advantage. On the other hand, simultaneous adoption by multiple operators may reduce collective profits due to competitive pressures [115]. This scenario is analogous to LH2 shipping. As, in this case, individual carriers may hesitate to implement advanced BOG management. Additionally, high upfront costs and market uncertainties are two other major challenges for shipping companies. Due to these, they would show less attraction to adopt high-efficiency propulsion, or optimized scheduling.

In summary, it can be concluded that LH2 shipping is not only challenged by its inherent technical constraints but also by the broader operational and economic environment. Port infrastructure limitations, long-haul voyage distances, fluctuating fuel costs, and carbon pricing policies can all materially affect shipping companies' decision making. Based on these, the companies shipping efficiency and profitability also fluctuate. Therefore, optimizing LH2 transportation requires an integrated approach that considers both technical performance of storage and BOG utilization systems. They would need practical operational factors among carriers. Overall, these factors include infrastructure readiness, route planning, regulatory frameworks, and cooperative strategies. The combined approach would help the shipping companies to mitigate challenges and gain profit efficiently.

9. Cooldown strategy and thermal management of LH2 carrier tanks

Studies in literature with considering the cool down strategy, management, and associated challenges of the cryogenic tanks of LH2 carriers are very limited. Wang et al. quote this demonstration in their study at 2024 [76]. The authors of this study have also not found any specific detailed study based on cool down management of LH2 cryogenic tanks. But there are sufficient studies to analyze the cool down challenges of LNG carrier tanks. In 2008, Lee et al. conducted scaled down experiments using 200 m^3 prismatic LNG tank model. In this experiment, liquid nitrogen was employed as a substitute cryogen to replicate the thermal behavior of LNG systems. The experiments simulated the initial cool down from ambient temperature to cryogenic conditions. Results showed a significant time lag between the temperature drop in the vapor region and the response of the insulation layer. It also highlighted that vapor-space cooling occurs much faster than heat removal from the tank walls and insulation [116]. Lu et al. (2016) performed numerical modeling of the cool down process in a large prismatic LNG tank [117]. They focused on coupled heat transfer and phase change processes. The simulation predicted limited vapor phase thermal stratification as spray cooling and evaporation effectively mixed the vapor. They demonstrated that the inner tank wall reached the liquid stratification temperature within approximately 2 h. This marked the end of the rapid cooling stage. After about 12 h of continuous cooling, a linear temperature gradient developed within the inner insulation layer. This indicated the establishment of quasi-steady conductive heat transfer [117]. The study provided valuable insights into transient wall and insulation temperature distributions. These are critical for assessing thermal stress, cool down time, and cryogenic efficiency. The findings also supported the design of optimized spray-cooling strategies. There are other studies on thermodynamic response in LNG storage tanks under static pressurization and sloshing conditions [118]. These studies solely investigated the thermodynamic and cool down characteristics of cryogenic tanks of LNG carriers. But due to the different thermal behavior of LH2, the findings from these studies cannot guarantee the cool down timing, heel management, or other thermodynamic estimations of cryogenic LH2 tanks.

To analyze cool down management of ship-borne cryogenic storage tanks of LH2 carrier during ballast voyage, Wang et al. (2025) first conducted a study comparing LNG vs LH2 ballast voyage strategies. They used analytical methods in MATLAB to build a thermal model for LH2 and LNG tanks [76]. They used this model from a previous study of them, which is described in the later part of the manuscript. The model included 2D insulation and lumped vapor/liquid masses. This model was used to resolve tank wall, insulation, liquid, and vapor interactions during the entire round trip (ballast + laden) voyage. Figure-9 represents the analytical lumped mass model of their study [76]. This provides a robust framework for simulating the heat and mass transfer within a cryogenic cargo tank. This approach involves two interconnected components. The first one is the phase discretization and stratification modeling (left panel). It involves two distinct lumped masses: one for the liquid and another one for the vapor. The vapor mass is further divided into vertical sub-layers. It is done to accurately capture thermal stratification phenomenon in the vapor space (ullage) which is a key factor for heat transfer and evaporation rates. The second component is the thermal resistance network (right panel) which represents the conductive and convective heat transfer through the tank's insulation system and walls. This ultimately solves the total heat flux into the liquid and vapor masses.

They considered different cool down strategies such as continuous vs. intermittent spraying and allowed tanks to warm up to different temperature limits. They also included the effect of tank wall and insulation thickness of tanks due to the wall being thicker in the case for LH2 tanks. They looked at heat transfer, residual energy, and boil-off generation during ballast voyage, pre-cool-down before reloading, and laden voyage. The outcomes of keeping tank cold vs. letting it warm up, and spray cool-down modes were compared [76]. Figure-10 illustrates the overall process of bulk carriage and unloading. It shows how the cargo is handled throughout a complete voyage cycle. It includes four phases of the total voyage. The first phase starts with laden voyage (full tank). The ship departs with fully loaded cryogenic tanks toward receiving terminal. It carries LNG or LH2. Then, comes the second phase: unloading process. At the destination, the liquid cargo is discharged to port storage facilities. The third phase is the ballast voyage which represents the return trip. The ship returns with nearly empty tanks. It contains only a small remained heel. During this stage, thermal stratification occurs due to the low fill level [120,121]. The upper vapor layers warm faster and increases the tank's internal temperature. The final phase is the pre-cool down before reloading. The remaining heel liquid at the third phase is sprayed inside the tanks through internal spray bars to cool down the tank walls. This is conducted before the

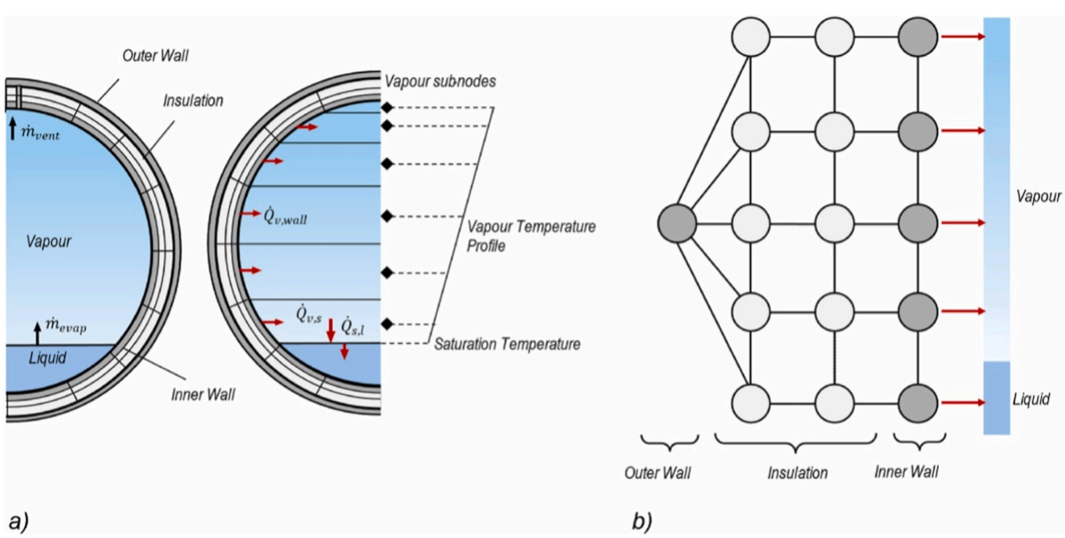

Figure-9. Schematic of the analytical lumped mass model for cryogenic tank thermodynamics. Figure (a) represents the discretization of the tank's vapor and liquid phases. It illustrates the lumped mass approach with vapor sub-layers and used to model thermal stratification. Figure (b) shows the corresponding thermal resistance network for the insulation and tank walls to calculate heat ingress to the system. The assumptions of this model include: considering axisymmetric conditions, and neglecting sloshing and para-ortho hydrogen conversion [76,119].

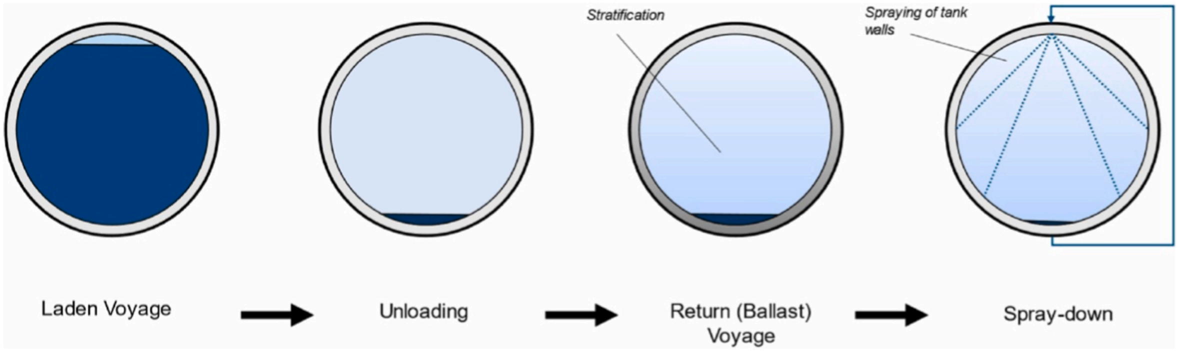

Figure-10. The schematic representation of the overall process of bulk carriage and unloading. It includes four phases: laden voyage (full tank), unloading process, ballast voyage (return trip), and pre-cool down before loading. Stratification occurs at the third phase when the ship is onboard with little heel in the tanks. Spraying of tank walls are conducted in the fourth phase [76].

arrival of the ship at the loading terminal. This ensures safe and efficient reloading of the next LH2 batch, and these 4 phases repeat again [76].

Figure-10 is added to understand the operational management process of a full LH2/LNG shipping cycle. It shows how temperature control and pre-cooling are managed between unloading and reloading.

The results demonstrated that allowing the tanks to warm up during ballast voyage helps reduce overall round-trip losses. This reduction is approximately 32-56.3% for LNG and 41.6-54.3% for LH2. This is because less energy is spent maintaining cryogenic conditions in empty tanks. But this increases excess boil-off during loading and laden voyage. This value was found up to 17% for LNG, and up to 5% for LH2. The results also exhibited that LH2 warms up more slowly than LNG due to the thicker insulation need of LH2 tanks along with heavier tank walls. Additionally, this slower warm up is also affected by hydrogen's higher heat capacity. Results indicated that inner tank wall thickness is a key factor in thermal response. They concluded that optimal strategy depends on voyage length, ship's BOG handling equipment (compressors, re-liquefaction), and fuel consumption profile [76].

These results of the reduction in heat accumulation associated with controlled tank warm-up are based on analytical thermal-mass transfer modeling. These values or results are not experimentally validated yet. So, the findings should therefore be interpreted as a modeling-based recommendation. Based on this, future work should focus on experimental testing or ship-scale measurements. This testing should include the verification of the predicted trade-offs between ballast-voyage heat accumulation and excess boil-off during loading and the laden voyage.

This study is the first approach to systematically compare ballast voyage management for both LNG and LH2 carriers. It significantly highlights some of the key design parameters and their contribution for thermodynamic performance of LH2 tanks. Key insights are related to how design parameters, such as membrane vs. Type-B tanks, insulation thickness, and wall thermal mass, influence cool down efficiency and BOG generation. One of the major challenges-finding balance between reducing ballast BOG losses and the resulting increase in excess BOG during loading plus next voyage is also examined in the study. This study presents the first approach where this unique and numerical trade-off is analyzed based on technical and operational perspectives.

Although this study demonstrates comprehensive approaches, fundamental results, and ways to overcome some key challenges of LH2 transportation, it offers some unresolved challenges and paves the way for future research directions. Several assumptions were considered while modeling the system. The major ones include negligible droplet-vapor interaction, lack of full scale experimental data on LH2 tank cool down, heel management, heat transfer behavior during ballast voyages, etc. It also assumes idealized spray impingement for spray cooling, and neglects droplet dynamics (size distribution, evaporation, vapor interaction, etc.). Additionally, in practical case heat transfer through support structures and ports occur, but this study ignores these kinds of secondary level heat transfer. But, the heat transfer amount through these structures are not too less to be ignored and can contribute significantly to total heat ingress. So, the accurate thermodynamic prediction needs all of these practical heat transfer ways to be analyzed for better tank performance and re-liquefaction load sizing. Moreover, this study assumes a certain heel mass and sprays frequencies for cooling objective, but does not optimize them or specify them for ultimate effectiveness. So, how much heel volume is actually needed for specific voyages to obtain minimum round trip losses remains an open practical and operational question. Lastly, the study does not couple the BOG utilization for propulsion or power generation systems. This coupling is important to analyze the performance strategy of thermal management.

In current LH2 transport systems, heel management practices are primarily limited to small-scale vessels. The Suiso Frontier employs a Type-C tank. It relies on maintaining a minimal liquid heel during ballast voyages to preserve cryogenic conditions. This also limits tank cooldown requirements prior to reloading. In this case, conservative operational control is followed for heel management. Active thermal optimization is not followed that much. And, excess boil-off gas is managed through controlled venting. Additionally, most advanced heel management strategies including controlled tank warm-up during ballast voyages and intermittent spry cooling before loading are still analytical or conceptual. Due to the absence of large-scale LH2 carriers, these strategies cannot be tested in real-world scenarios. As stated before, Wand et al. (2024) and Wang et al. (2025) numerically showed that such advanced strategies would reduce round-trip heat accumulation by approximately 41-54% compared to passive thermal management [27,54]. These approaches have not been experimentally implemented yet, but they provide the first quantitative framework for LH2-specific heel management. These studies also highlight the significant potential of applying these strategies in real-world large-scale LH2 carriers with systematic experimental studies, and omitting conservative strategies in current limited small-scale demonstration vessels. To the authors' best knowledge, there are some vessels currently under construction with advanced innovations as LH2 carriers where active heel management would be considered. These include Kawasaki 40,000 m^3 carrier (scheduled for sea trials by 2030), Samsung Heavy Industries LH2 carrier (aimed at commercialization by 2030), HD Hyundai LH2 carrier (scaling to 40, 000 m^3 by 2032). These informations are collected from online news. To the authors' findings, any dedicated research articles have not been found validating these information.

10. Thermodynamic modeling and stratification behavior at low fill levels in cryogenic LH2 tanks

10.1. Design fundamentals of LH2 tanks at low fill levels

When LH2 carriers have storage tanks with low fill levels, it becomes very challenging to accurately predict heat transfer, stratification, and pressurization behavior by simplified thermodynamic models. In the case of LH2 storing high-vacuum adiabatic low-pressure tanks are commonly used to minimize vapor losses. However, when the liquid occupies a smaller volume relative to the vapor space, even minor heat ingress can cause significant self-pressurization and BOG [122–125]. Figure-11 illustrates the fundamental design and structure of LH2 tank that is used for storage and transportation purposes [123]. It comprises an inner vessel which is the primary container that holds LH2. It effectively shows how a storage tank is specially designed to minimize BOG, which is the most significant challenge for both LH2 storage and transportation. The insulation layer is basically multi-layer insulation (MLI) to effectively restrict heat transfer. It is located between the inner vessel and the outer shell, and maintained under a high vacuum. Heat transfer through conduction, convection, and radiation are eliminated by this MLI. The structural supports are designed with mostly non-conductive materials to make those thermally inefficient and reduce heat transfer through these secondary parts. Additionally, unavoidable heat ingress causes excess hydrogen vapor which are vented by pressure relief device. The transfer pipe is used for loading and unloading the LH2. The grounding point prevents static charge accumulation and ensures safety [123].

At low fill levels, vapor volume increases and the rate of pressure rise accelerates. It is caused due to reduced thermal buffering of the liquid phase. This phenomenon emphasizes the need for accurate thermodynamic modeling. To further enhance storage performance and reduce BOG, cryo-compressed hydrogen is being explored [126–131]. This technology offers cryogenic storage and moderate pressure to achieve higher density. It combines the hybrid benefits and longer dormancy periods. Figure-12 illustrates the design of a cryogenic-compressed hydrogen storage system [131]. It is a hybrid technology that aims to overcome the key limitations of LH2 transport. The LH2 fill tube is used for filling the tank with LH2. The vacuum pump out port is part of the insulation system. It creates a vacuum around the inner vessel, thus minimizing heat ingress and boil-off. The GH2 fill tube allows for gaseous filling. The heat exchanger is a critical component that uses the cryogenic cold inside the tank. It cools the hydrogen gas being supplied to the engine and improves efficiency by managing temperature. Overall, this design represents cryogenic liquid handling, gas pressure management, integrated system operation, and critical safety systems, which are prerequisites for LH2 storage and transportation. Although this design was provided with an aim to LH2 storage, it gives valuable insights for LH2 transportation.

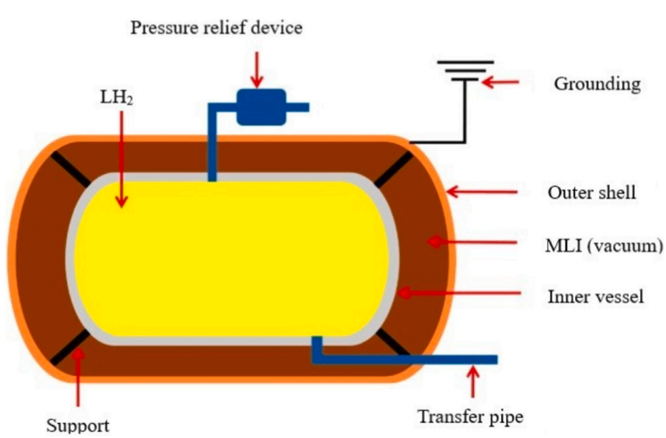

Figure-11. Schematic of a typical cryogenic LH2 storage tank. The tank is specially designed with a double-walled vacuum-insulated feature. There is an inner vessel that holds the cryogenic LH2. It is protected by vacuum multi-layer insulation (MLI)., andholds the outer layer surrounding it to reduce heat transfer. Pressure relief device and grounding point ensure operational features and critical safety [123].

10.2. Development of thermodynamic models and validation approaches

Although the studies conducted for LNG carriers give valuable insights towards LH2 transport, works on explicitly addressing heel carrying practices in LNG are scarce. Most of the studies available are operational or descriptive rather than analytical. So, the proper guide towards LH2 heel management and pertinent issues is also scarce [132]. Studies show that the Suiso Frontier is the only operational LH2 carrier. It has small-scale Type-C tank of 12.50 m^3, which is not representative of future large-scale commercial LH2 vessels [133–135]. So, the lack of design data and performance validation for large LH2 tanks design and thermodynamic analysis is significant. There is also the absence of standardized heat transfer correlations for cryogenic hydrogen. Wang et al. (2024) developed a thermodynamic non-equilibrium model to analyze how heat transfer, pressure rise, and BOG generation differ depending on different variables such as tank geometry, operating pressure, and initial liquid level [136]. They compared thermal performance among different tank shapes and investigated the effect of tank geometry on pressure rise rate, evaporation rate, and BOG losses. They demonstrated that non-equilibrium thermodynamic modeling better captures the temperature differences between liquid and vapor phase. It is different from equilibrium models that assume uniform temperature. In a previous study by Wang et al. (2023), they developed an analytical theoretical model to explain and quantify the heat distribution ratio (HDR) [137]. Their investigation specially focused on the physical mechanism behind how heat leaks differently into vapor and liquid due to the inner wall's thermal conduction, the distinct thermal properties of liquid and vapor hydrogen, and tank geometry and initial liquid fill ratio. They developed the HDR model to replace the conventional and less accurate uniform heat flux boundary assumptions. They analyzed the effect of different parameters such as fill level, total heat leakage, superheat, saturated pressure, and interfacial mass transfer rate on HDR. They coupled the new HDR model with a Thermal Multi-Zone Model (TMZM) to achieve high-accuracy pressure predictions [137]. This approach reduced pressure prediction error by over 60% compared to uniform flux models. Their study uniquely developed a theoretical model that quantifies how heat leakage is non-uniformly distributed between vapor and liquid in LH2 tanks.

Matveev et al. (2023) conducted a study to analyze how the size of LH2 storage tank affects self-pressurization and venting behavior. They used a simplified lumped-element thermodynamic model that treats the liquid and vapor inside the tank as two uniform zones with average temperature and pressure [138]. They used this model to simulate how heat leaks into the tank. This ultimately causes self-pressurization (when the tank is closed), and constant-pressure venting (when hydrogen is released to control pressure). They validated the model using NASA experimental data from the Multi-Purpose Hydrogen Test Bed (MHTB) tank [139] and applied it to tanks of different sizes (2-1200 m^3). The validation was performed against ground-based experimental data. Following this laboratory-scale validation, the model was applied in a parametric and scaling analysis to tanks of similar geometry with volumes. This allowed assessment of size-dependent pressurization rates, venting behavior, and boil-off losses. Figure-13 shows the geometry of MHTB tank. This tank is made of aluminum and also comparable to space applications with full-scale cryogenic tanks. The total tank height is 3.05 m. It has a vertical cylindrical insert of diameter 3.05 m and height 1.525 m. The tank's top and bottom surfaces resemble semi-elliptic caps, with a minor (vertical) axis that is twice as narrow and two (horizontal) axes that have the same diameter as the cylindrical section. A vacuum chamber holds the entire tank. The exterior of the tank is coated with a 45-layer insulation blanket made of Dacron netting and Mylar sheets, as well as a 1.4-cm-thick spray-on insulation. The experimental tank configuration also incorporates the control, transfer, and instrumentation systems [138].

Results showed that larger tanks experience slower pressure rise and lower relative BOG losses. The reduction was measured about 1.8% to

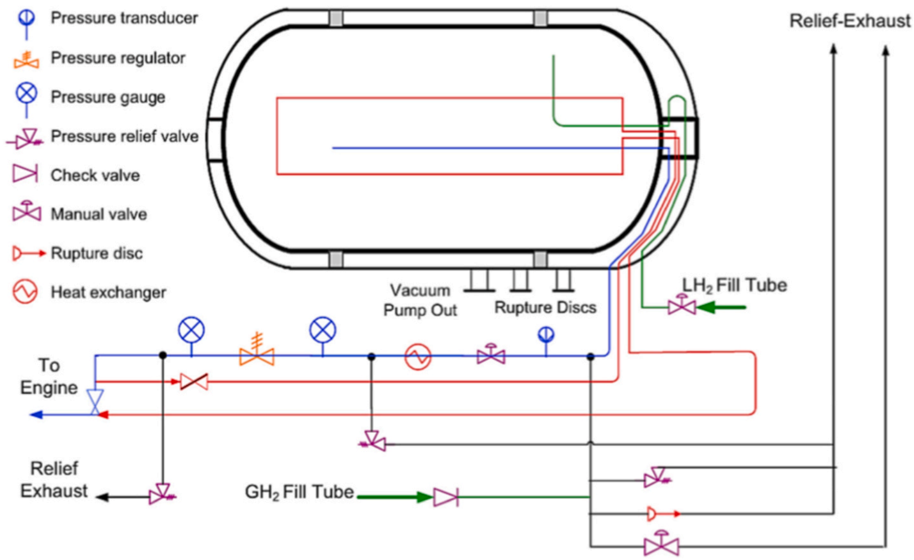

Figure-12. Schematic diagram of a cryo-compressed hydrogen storage system. It highlights the dual-function nature for handling both cryogenic liquid and high-pressure gas. It includes key features such as separate fill ports for LH2 and GH2, a heat exchanger, multiple safety devices, and vacuum insulation. The vacuum insulation is essential for maintaining cryogenic temperatures [131].

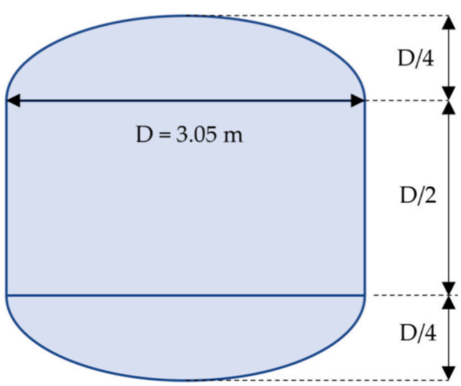

Figure-13. Geometry of Multi-Purpose Hydrogen Test Bed (MHTB) tank [138, 139]. This tank is comparable to a full-scale cryogenic tank. It was extensively used for experimentation by NASA and gives valuable insights for LH2 transportation by sea going vessels.

0.2% per day for 2-1200 m^3 tanks. They also demonstrated that BOG losses scale approximately with the inverse of tank diameter. The model's predictions agreed well with NASA's experimental data obtained from MHTB tank. Although these findings are derived from stationary storage analysis, are highly applicable for LH2 carriers or LH2 transportation. Because, in both cases BOG and maintaining pressure stability are two major challenges. However, the BOG and pressure pattern during transportation would vary due to the non-stationary feature and long-term exposure to surroundings.

It is important to clarify that the majority of thermodynamic models developed for LH2 storage and transportation have been validated against ground-based experimental facilities. Full-scale shipboard measurements do not exist at full scale yet. In particular, the validation of lumped-element and multi-zone self-pressurization models has primarily relied on laboratory-to-pilot scale cryogenic tanks. These are operated under controlled conditions, such as the NASA Multi-Purpose Hydrogen Test Bed (MHTB).

These studies give valuable insights for thermodynamic modeling of cryogenic tanks of LH2 carriers. But, in most of the cases they used inconsistent tuning parameters for natural convection heat transfer correlations. There is no consensus on accurate coefficients for vapor-liquid or wall-vapor heat exchange in self-pressurization modeling. Additionally, most existing models treat vapor and liquid as single nodes [140,141]. These models fail to capture vapor thermal gradients at low fill levels. This failure causes up to 60% under-prediction of wall internal energy, especially at ≤5% fill conditions [142]. Some studies introduced multi-layer vapor models, but their validation was limited to small laboratory-scale vertical cylindrical tanks [143,144]. Moreover, previous studies focused mainly on full-tank steady-state conditions. The transient thermal behavior when the tank is mostly vapor-filled, has not been experimentally validated [145–152].

To meet up the mentioned research gaps and analyze the heel required in LH2 transportation, and heat transfer and stratification behavior in LH2 tanks compared to LNG tanks, Wang et al. (2024) developed a new thermodynamic analytical model [142]. Their objective was to predict heat transfer into LH2 tanks, vapor stratification at low fill levels, and the effect of tank pressurization and insulation type. Their study focused on a 21-day ballast voyage scenario. They studied how different insulation types perform to reduce heat ingress. They especially compared LH2 spherical tanks with perlite or polyurethane foam (PUF) to LNG membrane tanks; as practices are mature for LNG but not for LH2.

In their study they mentioned one limitation of single-node models. Single node-models can only return a single mass-averaged vapor temperature. At low fill levels, where substantial vapor thermal stratification may be anticipated, this presents a problem. If the vapor is treated as uniform in temperature at the mass-averaged temperature, the internal energy of the wall will be underestimated for a partially filled tank with a linear temperature distribution in the ullage (sub-portion a; Fig. 14) and wall temperatures equal to the adjacent vapor and liquid. Sub-portion b of Figure-14 quantifies the consequence of the simplification. It shows that when the hot vapor with a maximum temperature over 100 K exists, the simple single-node model significantly under-predicts the internal energy which is stored in the tank's steel wall. The wall in contact with the hot top vapor gets much warmer. Due to that, the energy required to heat steel increases non-linearity at cryogenic temperatures. Prediction errors may surpass 60% at a fill level of 5% when maximum vapor temperatures surpass around 100 K. This is explained by the fact that the specific heat of metals like stainless steel is highly sensitive to temperatures in the cryogenic range [142].

They considered a double-walled cryogenic tank and it was initially close to 100% fill level under constant pressure venting, representing by

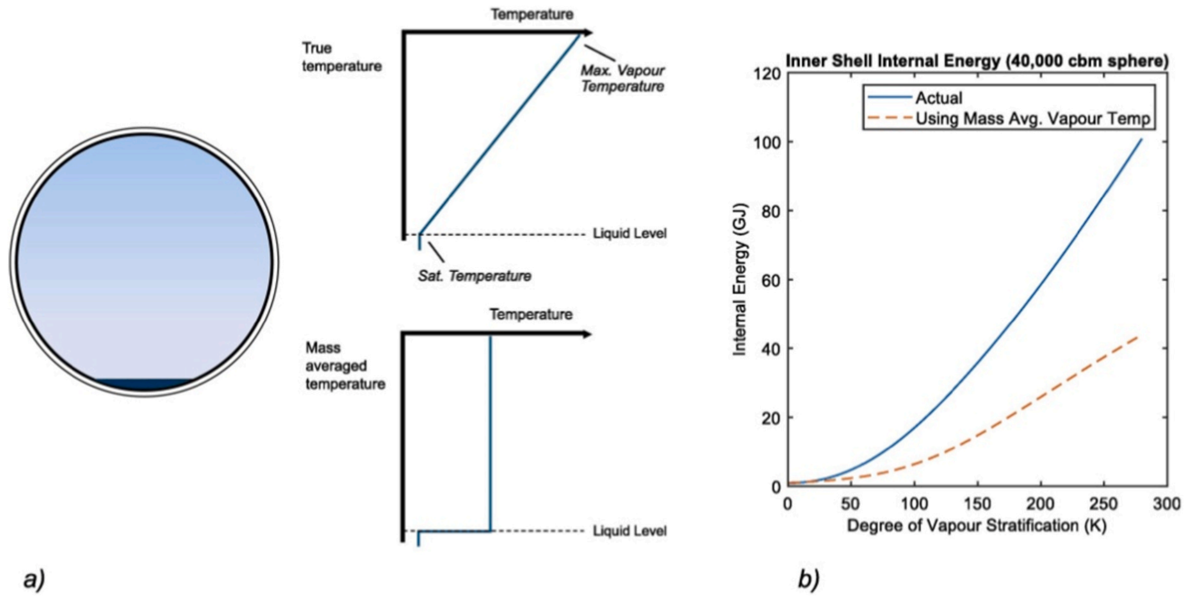

Figure-14. Impact of vapor thermal stratification on tank energy calculations at low fill level. (a) Represents the comparison between a realistic, stratified vapor temperature profile against a simplified single-node model's mass-averaged temperature. (b) Shows the resulting under-prediction of the inner shell's internal energy. It demonstrates a critical error that exceeds 60% at high stratification [142].

Figure-15. They assumed the inner shell temperature to be uniform and equal to the saturation temperature of the liquid for simplicity. They also assumed the tank to be in a quasi-steady state condition [142]. This figure helps to understand where the heat comes from and how it moves through the tank walls. It also shows how the heat ingress leads to BOG, especially after the cargo is unloaded.

Due to the constraints of single-node model, they proposed an analytical model which is represented by Figure-9 [76]. The explanation of the figure is presented in Section-9.

10.3. Comparative heat transfer analysis for LH2 and LNG transportation systems

Wang et al. accumulated the analysis of the heat buildup in near-empty LH2 and LNG tanks during ballast voyage (return trip) in Figure-16 [142]. The figure represents the variation in $\frac{\dot{Q}_{environmental}}{Q_{steady\ state,f}}$, $\frac{Q_{residual}}{\dot{Q}_{environmental}}$, θ_1, and θ_2 over 21 days for both LH2 and LNG cases at 5% heel under atmospheric venting. U represents internal energy. θ_1, θ_2, and $U_{residual}$ are represented by Equation (1), Equation (2), and Equation (3) respectively [142].

$$\theta_1 = \frac{\int_0^t Q_{environmental} dt}{Q_{steady\ state,f} \times t} \tag{1}$$

$$\theta_2 = \frac{U_{residual}}{\int_0^t Q_{environment} dt} \tag{2}$$

$$U_{residual} = U_{tank} - U_{tank,steady\ state,f} \tag{3}$$

The figure shows that the overall environmental heat transfer into the tank remained nearly constant during the simulation period. It indicates that only a limited reduction in heat load happened due to allowing the tank to gradually warm up. Specially, the environmental heat transfer for perlite-insulated LH2 tank decreased by only 3.6% after 21 days. But, the cumulative heat transfer was 2.3% lower compared to steady-state conditions at 5% fill. The reduction in instantaneous and cumulative heat transfer for the polyurethane foam (PUF)-insulated LH2

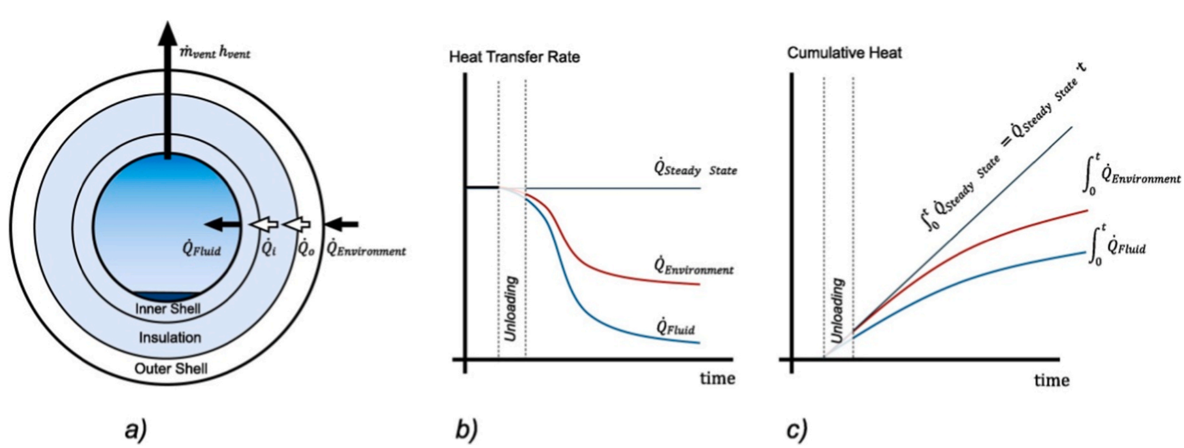

Figure-15. (a) Diagram of heat flow path within a double walled tank. Environmental heat constantly flows to the outer shell. Then passes through insulation, inner shell, and finally enters the liquid hydrogen and creates BOG. Figure-(b) shows the heat pulse after unloading. The heat that was going into the liquid suddenly has nowhere to go and causes a pulse of heat to build up inside the tank's structure. Figure-(c) shows the total heat stored in the tank walls cumulatively. When new, cold LH2 is loaded, this heat is released into it. This causes a large and immediate burst of BOG [142].

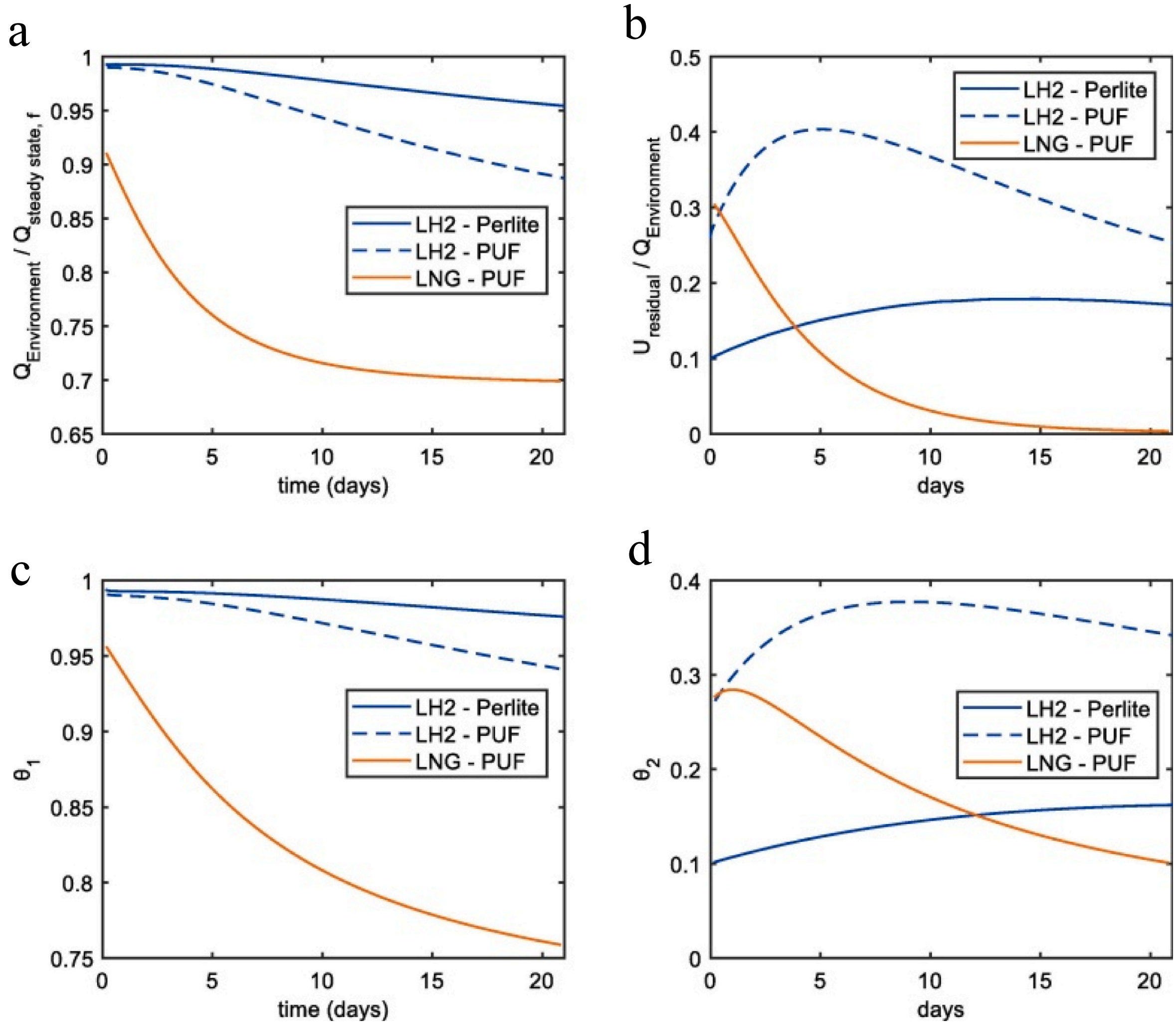

Figure-16. Variation of thermal behavior in cryogenic storage tanks at 5% heel. These are measured under atmospheric venting conditions. (a)illustrates the ratio of instantaneous environmental heat transfer to its steady-state value. (b)represents the ratio of tank heat gain to environmental heat transfer. (c)shows the dimensionless cumulative environmental heat transfer, while (d)depicts the residual heat within the tank wall. The results correspond to a 21-day simulation period following ship departure. It highlights the comparative thermal responses of LH2 and LNG tanks during low-fill operation [142].

tank is slightly higher, 8.7% and 5.7% respectively over the same period. These findings demonstrate that the benefit of reduced environmental heat transfer through tank warming is minimal. This is also true for the case of storage at low fill levels over several weeks. In contrast, the warming effect is more significant for the case of LNG carrier. The PUF-insulated LNG tank experiences a 24% reduction in cumulative environmental heat transfer after 21 days. The non-independent LNG tank has a relatively thin inner wall. It stores only 7.9% of the total thermal energy. This allows it to warm up more quickly and thus reduce heat ingress from the environment. In comparison, the LH2 tank's inner wall contained about 76.6% of the total stored heat energy [142]. This results in slower wall temperature rise and smaller reduction in heat transfer during same period.

They concluded that the LNG tanks demonstrated a tendency to reach a quasi-steady thermal state within approximately 3 weeks. In this state, the heat transfer from the environment to the outer shell and from the inner wall to the internal fluid became balanced. Conversely, the LH2 tanks continued to accumulate heat throughout the same period. This reflected their higher insulation thickness and lower thermal conductivity [142]. This continuous heat buildup in LH2 tanks emphasizes the need for careful design considerations. As proper design is mandatory to prevent excessive pressure rise or BOG losses during extended low-fill operation.

Overall, this analysis highlighted that self-warming during low-fill storage provides limited thermal relief on LH2 tanks than LNG systems. This finding opens the pathway for precise thermodynamic modeling. It also emphasizes effective insulation strategies when operating LH2 tanks at low fill levels.

10.4. Long-term performance, aging, and lifecycle considerations of LH2 marine insulation systems

Unlike stationary onshore storage, shipborne LH2 tanks are continuously exposed to dynamic mechanical loads. They also face cyclic pressurization, sloshing-induced stresses, vibration, and a corrosive salt-laden environment. All of these can accelerate insulation degradation over the vessel lifetime. Vacuum-based insulation systems, such as evacuated perlite and multi-layer insulation (MLI), offer superior thermal performance at the beginning of service life. But their effectiveness is highly sensitive to vacuum integrity. Over extended operational periods, gradual vacuum degradation may occur. There are various reasons behind this. The most significant ones are: permeation, micro-leakage, seal aging, and outgassing of residual gases. These reasons are prominent especially during repeated warm-up and cool-down cycles associated with ballast voyages. Even modest increase in annular pressure can lead to a significant rise in effective thermal conductivity.

This increases heat ingress, self-pressurization rates, and cumulative boil-off losses. So, periodic monitoring, re-evacuation, or insulation refurbishment is required to maintain long-term vacuum performance. However, these would introduce additional operational complexity and lifecycle costs [142,153].

In contrast, non-vacuum insulation systems such as PUF show lower sensitivity to vacuum loss. But they are more susceptible to aging mechanisms. These are related to moisture ingress, thermal cycling, and mechanical fatigue. Under marine conditions, repeated exposure to temperature gradients, vibration, and structural deformation can induce microcracking and degradation of the foam matrix. These would create a gradual increase in thermal conductivity over time. Moisture absorption further intensifies this effect. Particularly in humid maritime environments. This leads to irreversible deterioration of insulation performance and increases boil-off rates [142,153].

From a lifecycle perspective, insulation aging directly influences some major factors. These include long-term BOG generation, energy losses, and the effective cost of transported hydrogen. Short-term thermo-economic models typically assume constant insulation performance. But long-term degradation may shift the relative competitiveness of insulation concepts. This is especially prominent for long-haul routes and high-utilization vessels. Consequently, insulation selection represents not only a thermal design choice but also a strategic decision. These would certainly affect maintenance schedules, vessel downtime, and lifetime operating expenditure [142,153].

10.5. Applicability and validation of reduced-order and multi-zone thermodynamic models

Based on the discussions on single-node and multi-zone models in Section-9 and Section-10, several comments can be provided on the improvement of multi-zone models with real-world data to demonstrate more practicality. According to analysis, current multi-zone and thermal stratification models for LH2 tanks remain largely analytical. The scarcity of full-scale experimental data has made its deployment feasibility less practical. Current multi-zone models show potential predictive capability in practice only by significantly improving through validation against real-world measurements. Different sectors of validation, such as spatially resolved wall temperatures, vapor temperature gradients, and transient boil-off rates during ballast voyages and pre-cooldown operations, need to be conducted. Data from instrumented large-scale LH2 tanks (although very limited) would allow to standardize and fix some of the core issues like the refinement of interfacial heat transfer coefficients, vapor-wall convection correlations, and stratification decay rates. Some of the key datasets need to be collected by distributed temperature sensors along tank walls, vapor ullage thermocouples, and heat flux measurements across insulation layers. Additionally, major ship-scale operational data would enable calibration of model assumptions. These assumptions are related to spray impingement, mixing efficiency, and transient heat storage within tank structures. Spray cooling effectiveness, heel distribution, and boil-off compressor loading profiles are some of the major operational data in this case. Such datasets are currently unavailable for commercial-scale LH2 carriers.

Moreover, discussions can also be conducted on the area of acceptance of reduced-order models and multi-zone models. Based on the understanding of model analyses, reduced-order single-node equilibrium models are most likely to be acceptable in situations where tanks operate near steady-state conditions, with high fill levels, limited vapor space, and weak thermal stratification. Under such conditions, vapor and liquid temperatures are relatively uniform. And, lumped-parameter approaches can also provide important estimates of average pressure rise and boil-off rates with minimal computational cost. This makes single-node models suitable for preliminary design studies. It is also suitable for parametric sensitivity analyses, and operational assessments where detailed spatial resolution is not required. However, multi-zone or stratified models are likely to be essential when LH2 tanks operate at low fill levels. The cases of ballast voyages, cooldown periods, or extended dormancy represent low fill levels phenomena. In these cases, strong vapor thermal stratification develops. This leads to non-uniform wall temperatures and significant heat storage within the tank structure. Single-node models are unable to capture these gradients. Previous discussions demonstrate that they can under-predict wall internal energy by more than 60% at fill levels below approximately 5%. This even becomes more significant when upper vapor temperature exceed 100 K. In summary, multi-zone models are therefore critical for accurately predicting transient heat accumulation, excess boil-off during reloading. There applications is also needed to perfectly assess cooldown requirements, and the resulting loads on boil-off gas handling systems.

10.6. Ship-scale integration constraints of spherical cryogenic tanks

Spherical cryogenic tanks offer superior intrinsic thermal performance due to their minimal surface-area-to-volume ratio. But this advantage does not offer that much impact while considering ship-scale structural, stability, and operational constraints. Studies on structural trade of spherical LH2 tanks demonstrate that vacuum-jacketed double-wall configurations require independent load-bearing outer shells. These are designed against external atmospheric pressure. Due to this, extra structural features, such as stability-driven stiffening, increased shell thickness, and additional structural mass, need to be considered [154]. These additions increase the overall thermal inertia of the tank system. They also introduce extra conductive heat paths through supports, penetrations, and attachment points. All of these contribute to passive heat ingress beyond that predicted idealized thermal models.

At the ship-integration level, spherical tanks are typically deck-mounted or semi-deck-mounted. This is especially observed in both LH2 demonstrator vessels and LNG MOSS-type carriers. This arrangement significantly reduces volumetric packing efficiency, and results in unused hull volume and lower effective cargo density at the vessel scale [155]. Additionally, the elevated vertical position of spherical tanks raises the ship's center of gravity (CoG). This elevation brings the necessity to increased ballast and hull reinforcement to maintain intact and damage stability. These stability corrections subsequently increase displacement and wetted surface area. Based on the propulsion basics, higher displacement and bigger wetted surface area increases the total resistance and propulsion power demand [156]. These lead to higher fuel consumption and indirect operational energy penalties.

Spherical tanks require additional structural supports, saddles, skirts, and deck foundations. These further increases parasitic loads and introduce thermal bridges that bypass insulation layers and contribute to steady passive heat losses. Experience from LNG MOSS carriers shows that these system-level penalties partially offset the low boil-off advantage of spherical geometry [156]. This is truer for large-capacity vessels where deck space utilization and longitudinal weight distribution become one of the most crucial design factors. These ship-scale integration challenges are true for both LNG and LH2 carriers. This critique highlights that the thermodynamic superiority of spherical tanks must be evaluated at the whole-ship level rather than at the isolated tank level.

In summary, spherical LH2 tanks minimize local heat ingress, but their requirement for heavier structural systems, increased ballast, center of gravity and center of buoyancy elevation, reduced free deck space, and higher propulsion energy demand substantially alters their potential performance advantage. These are true for both small-demonstrate level and large-scale realistic marine operating conditions.

11. Transferability of LNG heel management practices to LH2 transportation systems

After analyzing heel management and thermodynamic behavior of LH2 carriers in parallel with LNG carriers, one key question arises: which practices and strategies of LNG carriers are transferable to LH2 carriers, as they have different temperature and pressure limits. LNG carrier operations provide valuable operational insights of maritime energy transportation. But all LNG heel management practices are not transferable to LH2 systems because of their different fundamental thermophysical differences. Some practices are largely transferable. These include maintaining a minimum heel to avoid full tank warm-up, and intermittent spray cooling prior to loading. Moreover, recognizing the coupled nature of ballast and laden voyage can also be a transferable area. These principles are rooted in general cryogenic thermodynamics and remain valid for both LNG and LH2 carriers.

However, several LNG practices are not directly transferable; rather they would need significant modification for LH2. While LNG tanks tolerate higher wall temperatures and higher MAWP, LH2 tanks operate at much lower temperature (20 K). LH2 tanks also go through stricter pressure constraints. Due to these major mismatches, aggressive spray cooling strategies, rapid cool-down rates, and assumptions of rapid wall thermal equilibrium used in LNG are not directly applicable to LH2. Moreover, LNG membrane tanks have lower wall thermal mass. This allows faster warm-up and cool-down. In contrast, LH2 tanks exhibit significant thermal inertial due to thicker walls and insulation.

Several LNG practices are fundamentally unsuitable for LH2 systems. Allowing tanks to approach near-ambient temperatures during ballast voyages, relying on high pressure operation to suppress BOG, neglecting vapor stratification effects are some of the fundamentally non-transferable practices. Such practices would lead to excessive heat storage in the tank structure for LH2 carriers. They would also lead to severe BOG surges during reloading. All of these phenomena are demonstrated by recent analytical studies.

Based on these, cooling intensity, duration, and spatial distribution must be actively controlled and significantly reduced in the case of LH2. Any LNG-derived warming or cool-down practice of LNG should be paired with tight pressure control. LNG assumptions of rapid wall equilibrium need to be efficiently replaced with time-dependent, inertia-aware thermal models.

12. Sloshing-coupled thermodynamics in LH2 maritime transport

Ships face six degrees of motion in real-sea. These motions create sloshing in LH2 tanks. The variation in stratification, rapid fluctuations of internal vapor pressure, and heat transfer behavior changes with sloshing phenomenon. During long voyages, steady heat ingress through tank insulation is the major driving factor of the long-term average boil-off rate. Recent studies show that sloshing can introduce short-term thermodynamic effects of comparable importance in LH2 tanks. Phase change based CFD studies demonstrate that sloshing-induced interface deformation and repeated liquid-wall contact can significantly increase evaporation rates. This ultimately causes pressure fluctuations. But, to what scale these increased evaporation phenomena and pressure fluctuations would occur depend strongly on excitation amplitude, frequency, and liquid fill level [157].

Numerical investigations of LH2 tanks under sinusoidal sloshing indicate that evaporation increases with acceleration amplitude and decreases with increasing sloshing frequency. This acceleration is the imposed acceleration of the LH2 tanks due to 6 DoF external motion of the vessel. At the same time, higher storage pressure reduces pressurization rates [158]. Considering realistic ship motion with conventional excitation levels, the same study also reported that daily evaporation rates range from approximately 2.6% to 5.1% [158]. This highest margin of 5.1% represents a larger boil-off loss than the conventional boil-off loss measured without considering sloshing impact in the study of Al-Breki and Bicer (2020) [111]. These values suggest that sloshing induced evaporation can locally match or exceed the contribution from steady conductive heat ingress over similar time periods [158].

Studies further explain the high sensitivity of LH2 to sloshing by its fundamental fluid properties [149]. Due to the extremely low viscosity, high wettability, and low latent heat of LH2 compared to other cryogenic fuels, the thermal buffering capacity of LH2 is reduced. This enhances heat and mass transfer during interface motion [159]. As a result, even short sloshing events can disrupt thermal stratification. It can also generate pressure oscillating, and increase localized evaporation during long-voyages.

At the ship scale, coupled thermodynamic-motion models show that LH2 carriers experience boil-off rates nearly nine times higher than LNG carriers with similar insulation [160]. The most valuable finding in the case of sloshing effect is that the LH2 boil-off is approximately twice as sensitive to sloshing effects under changing sea states [160]. Moreover, industry-led benchmark studies confirm that static heat ingress models are insufficient under dynamic sloshing conditions. They also highlight the need to explicitly account for coupled sloshing, heat transfer, and phase change in large-scale LH2 tanks [161].

13. Future research directions

The following potential future research can be conducted based on ship-borne LH2 transportation.

a. Insulation is one of the most critical factors for tanks of LH2 carriers. Although, recent studies have taken into concern the impact of proper insulation and efficiency analysis, proper insulation thickness and material have still not been specified. Future works need to be conducted investigating the effect of different insulation thicknesses with different insulation materials and how they will respond to the efficiency of heat ingress reduction.
b. Spraying by cold LH2 before loading to maintain low heat gain in the tank has been proposed to reduce BOG. Intermittent spraying can be an effective method in this case. The amount of spraying, the duration of spraying, and spraying patterns such as continuous or discrete spraying can be investigated to find out the optimal spraying techniques. Although the pattern will vary based on the LH2 tank size, surrounding temperature, voyage duration, and tank material etc., a database can be produced for different patterns and variables by continuous studies.
c. Tank size is a critical factor for LH2 carriers as ships undergo space constraints. Additionally, larger carriers increase the deadweight of the ship which affects resistance and fuel consumption for propulsion. Moreover, heavy structural design needs to be considered to uphold the huge tanks. So, optimizing tanks size and shape is one of the most crucial factors. More experimental data on thermodynamic efficiency across a broader range of tank sizes need to be investigated.
d. The studies in Section-12 indicate that sloshing-induced thermodynamic effects are intermittent, but their cumulative impact during real maritime operation is significant. So, accurately estimating total boil-off requires real-world sloshing phenomena to be integrated. Fully coupled hydrodynamic-thermodynamic models need to be validated under representative ship motions. This sloshing-coupled thermodynamics is a key research gap for LH2 maritime transport.
e. Optimizing fill level is another crucial concern. Although studies have been conducted based on fill level effect and different scenarios with different amount of fill levels, the optimized findings of fill level is still an underdeveloped sector. How much fill level is feasible to get better techno-economical advantage is a potential research area. Again, the variation of fill level depending on potential BOG generation and self-pressurization due to variable voyage duration, tank geometry, tank material and insulation, and surrounding temperature etc. needs to be analyzed to better design of tank models and analyze whole transportation scenario.
f. Maritime LH2 transportation includes thermodynamics, operation, and economics. So, integrated ship system studies that combine storage thermodynamics, powertrain options including MGO, BOG and fuel cells, and voyage economics need to be conducted. Realistic

uncertainty in hydrogen production cost and carbon pricing should also be considered.

g. Practical methods for BOG management should be studied. It needs to include the analysis ranging from passive design (tank sizing/ insulation) to active systems (re-liquefaction, BOG-to-fuel systems, optimized heel and pre-cool strategies). These also need demonstration at representative scales.
h. Specifically, studies are limited for describing ship capital costs in terms of tank MAWP or insulation type due to a lack of publicly available data. A broader range of tank MAWP and tank insulation techniques may be the focus of future research. Additionally most of the studies are based on number of assumptions. The assumption should be specified for practical research output.
i. LH2 carriers can integrate dynamic wave energy conversion technologies which can be a significant research area. Insights from the studies based on WEC can be applied to develop hull-integrated or bow-mounted oscillating water column modules [162]. These would harvest wave-induced pneumatic energy during navigation. Such systems could also provide auxiliary renewable power for re-liquefaction units, BOG compression, navigation loads, and onboard hydrogen processing equipment. Research can be conducted on the feasibility of this system to investigate how ship motion, hull geometry, and variable sea states influence oscillating water column aerodynamic performance. Specially, it is needed to analyze structural integration when coupled with cryogenic hydrogen systems.
j. Waste heat recovery from propulsion engines or hydrogen fuelled power systems offer promising opportunity for future research. Towhid et al. (2025) demonstrated waste heat driven onboard desalination [163]. The same principles can be applied to LH2 carriers where significant low-to medium grade waste heat is available and can be used for cryogenic tank operations. Future research should explore the close-loop water-energy system onboard LH2 carriers where desalinated water could be used for electrolysis feedstock at receiving terminals, cooling loops, or tank maintenance operations.

14. Conclusion

Liquid hydrogen (LH2) is increasingly recognized as an acceptable and efficient energy carrier. It is significant for enabling long-distance and large-scale energy transportation to achieve a future world with net zero emission. LH2 carrier technology has significant strategic interest, but it remains in an early development stage. The technical readiness level (TRL) of LH2 carriers is substantially lower than that of LNG and other liquid energy carriers. This review consolidates the current state-of-the-art developments on LH2 transportation research. It includes thermodynamic behavior, boil-off gas (BOG) management, tank design, cool-down strategies, optimized model build-up, and techno-economic considerations related to ship-borne LH2 transportation. All these sectors have been analyzed and accumulated to identify key scientific and engineering bottlenecks.

Different transportation techniques such as LNG, ammonia, LOHCs, DME, and methanol have their own advantages. But LH2 surpasses these technologies by its superior gravimetric density. However, it also suffers from the lowest volumetric energy density and the highest refrigeration demand. The main challenge offers the largest BOG losses driven by extreme cryogenic conditions (-253 °C). It is evident that LH2 transportation offers the highest BOG losses among major liquid carriers. During long-distance ocean shipments, this loss could become around 3.44% per day. Studies based on the commercial analysis reveals that ship-borne LH2 transportation technologies are significantly underdeveloped. Currently, only short-distance pilot operations and a single small-scale LH2 carrier (Suiso Frontier) is available commercially. It indicates the huge lack of operational and experimental datasets which are responsible for persistent uncertainty in model validation and techno-economic feasibility.

LH2 transportation faces different unique and critical thermodynamic phenomena. Recent advancements in thermodynamic modeling have begun to address those issues. Non-equilibrium, multi-zone models demonstrate that vapor-phase thermal stratification at low fill levels can lead to >60% under-prediction of internal wall thermal energy. This happens when simplified single-node models are used. This under-prediction affects crucial estimations such as BOG estimation, pressure rise prediction, and re-liquefaction system sizing. Studies show that LH2 tanks store almost 76% of accumulated heat in the inner wall during ballast voyages and do not reach to a quasi-steady condition. Whereas, LNG offers only around 7.9% heat accumulation in these cases. This higher percentage of LH2 compared to LNG causes persistent heat ingress and delayed thermal stabilization. Studies show that if tank is allowed to warm during ballast voyage, round-trip losses can be reduced by a high amount of percentage (42-54%). At the same time this technique introduces substantial excess BOG generation during both loading and the subsequent laden voyage, which limits the potential of this technique. It also indicates to perform integrated thermal management rather than optional approach for efficient LH2 transportation.

The results from studies with different comparative modeling of insulation types confirm the limitation of insufficient thermal relief by current technologies. This is more acute during low-fill level conditions. This drawback suggests the importance to conduct study for the optimization of insulation technology including insulation thickness, structural heat-leak sources, and tank material thermal conductivity. Moreover, although some studies have dealt with heel volume management, but their optimization is still an unresolved area. This heel management optimization is needed as this factor is crucial for the key thermodynamic and operational performance, such as pressure stability inside tank, spray-cooling requirements, BOG recovery potential, etc. Moreover, several complex performance variables were neglected in previous studies. These include para-ortho conversion heat release, sloshing-induced convection, droplet-vapor interactions during spray cooling, heat ingress through support structures, dynamic voyage variations etc., all of which are crucial in real-sea transportation conditions.

In the LH2 supply chain, transportation remains the single largest cost element. This conclusion is demonstrated from techno-economic assessments. Different strategies have been taken for BOG optimization, but still LH2 shipping exhibits the highest cost per delivered energy among major carriers. LH2 shipping yields a relatively high transportation cost of around 3.74 $/GJ. This high cost is driven by high insulation demand, parasitic cooling loads, and BOG-dominant energy losses. The feasibility of LH2 transportation from the tech-economic perspective depends on some fluctuating and unresolved phenomena, such as strong carbon pricing frameworks, mature liquefaction infrastructure, significant improvements in large-scale carrier design with optimized MAWP, tank sizing, propulsion hybridization with BOG or fuel cells, etc.

Non-equilibrium stratification modeling, heat distribution ratio (HDR) formulations, and thermal multi-zone frameworks are some of the recent research advances in the LH2 transportation field. These advancements have significantly reduced computational inaccuracy. These have also improved the predictive understanding of transient thermodynamics in LH2 tanks. Experiment-informed modeling indicates that larger tank diameters can cut relative BOG losses from around 1.8% to around 0.2% per day due to thermal inertia. This is validated against NASA's MHTB tank data.

Overall, this review demonstrates that the transition from LNG-based design towards LH2-specific design is a time-demanded approach for ensuring zero-emission goal. This is also a fundamental scientific requirements based on complex engineering principles. The optimization of maritime LH2 transportation needs more fundamental studies which would include high-fidelity heat transfer correlations. For successful implementation, full scale experiments with considering sloshing-inclusive transient models need to be performed. These are more important for partially filled tanks which are the major center of focus for all kinds of analysis. This technology needs integrated ship-energy systems which can be achieved by the accumulated study of thermodynamics, propulsion, and re-liquefaction for commercialization. Several key basics for holistic LH2 transportation study and development, such as optimized cool-down strategy, heel management, and insulation effects over varying voyage durations, need to be optimized further. For successfully deploying maritime LH2 transportation, shared industrial data and standardized performance metrics need to be accumulated. Addressing these research gaps and working with the potential optimization technique, ship-borne LH2 transportation technology can turn into a cornerstone of the global renewable energy supply chain.

CRediT author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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